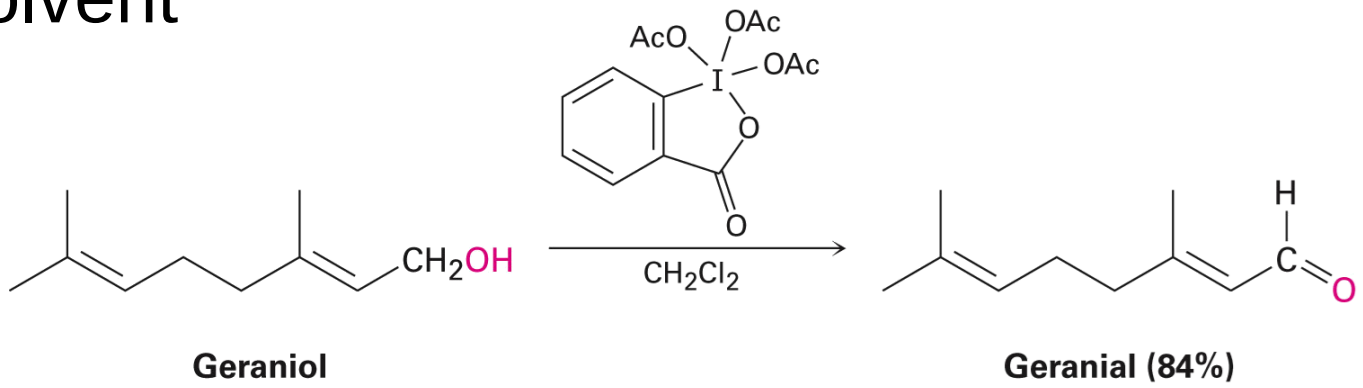
Chapter 19
Aldehydes and Ketones: Nucleophilic Addition Reactions

Preparing Aldehydes

- Oxidization of primary alcohols using Dess-Martin pyridinium reagent in dichloromethane solvent



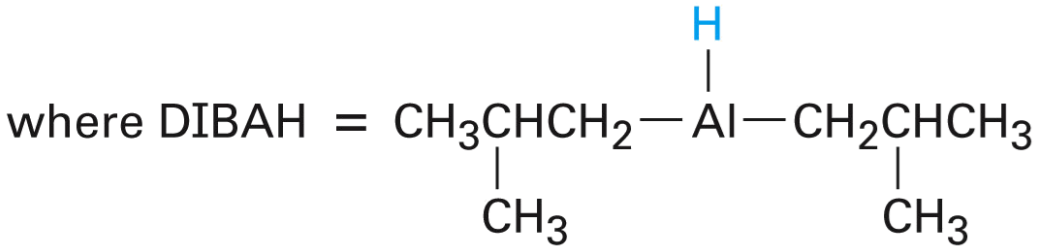
- Certain carboxylic acid derivatives can be partially reduced to yield aldehydes

Preparing Aldehydes

- Example
  - Partial reduction of an ester by diisobutylaluminum hydride

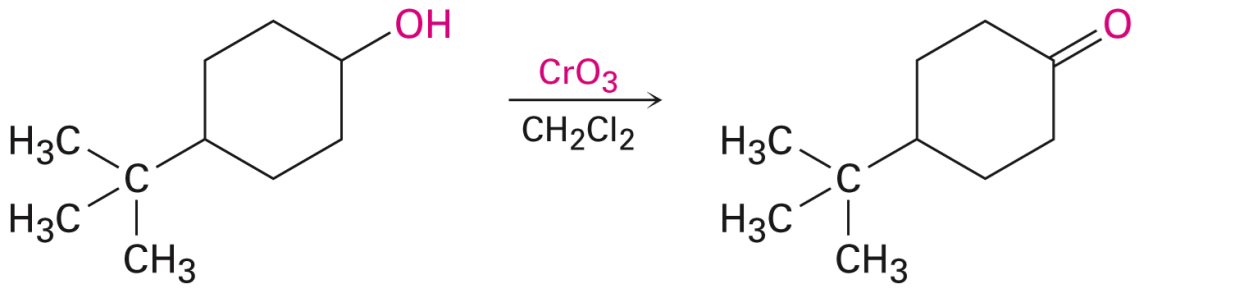


Methyl dodecanoate **Dodecanal (88%)**



Preparing Ketones

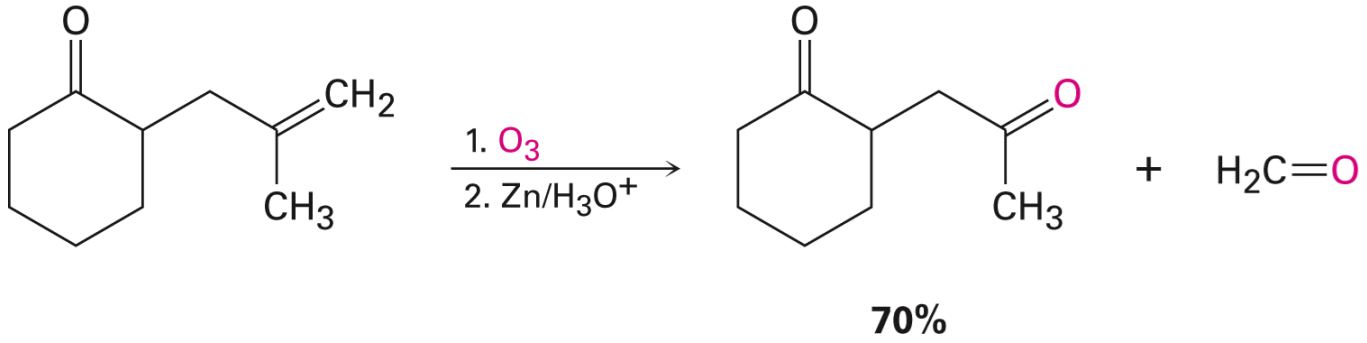
- Oxidization of a secondary alcohol
- Choice of oxidant is based on factors such as:
  - Scale
  - Cost
  - Acid/base sensitivity of the alcohol
- Dess–Martin periodinane or a Cr(VI) reagent are a common choice



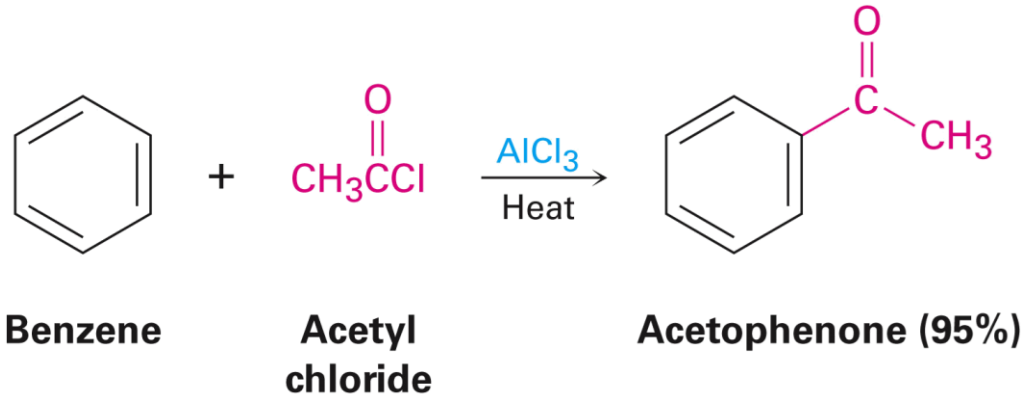
4-*tert*-Butylcyclohexanol **4-*tert*-Butylcyclohexanone (90%)**

Preparing Ketones

- Ozonolysis of alkenes yields ketones if one of the unsaturated carbon atoms is disubstituted

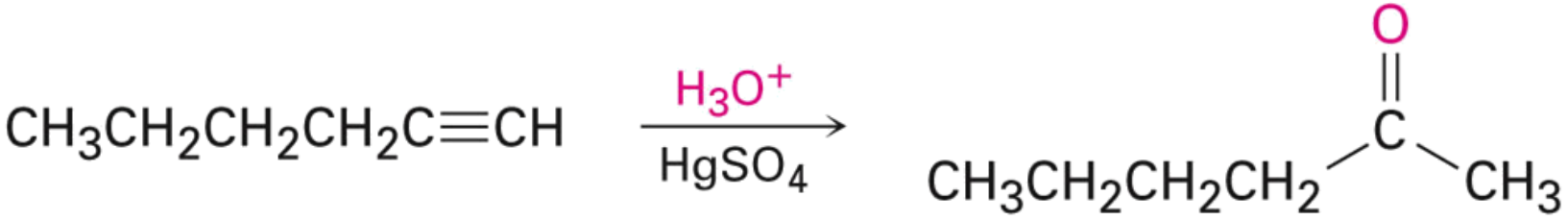


- Friedel-Crafts acylation of an aromatic ring with an acid chloride in the presence of $AlCl_3$ catalyst



Preparing Ketones

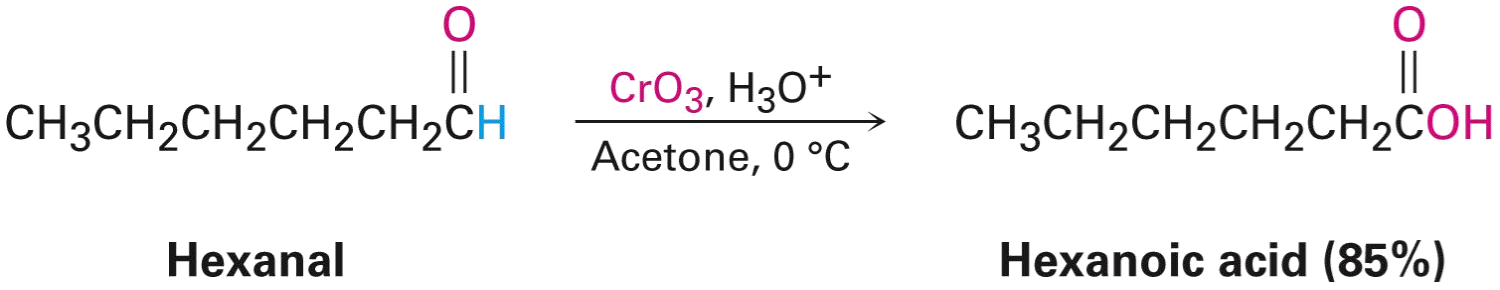
- Ketones can also be prepared from certain carboxylic acid derivatives



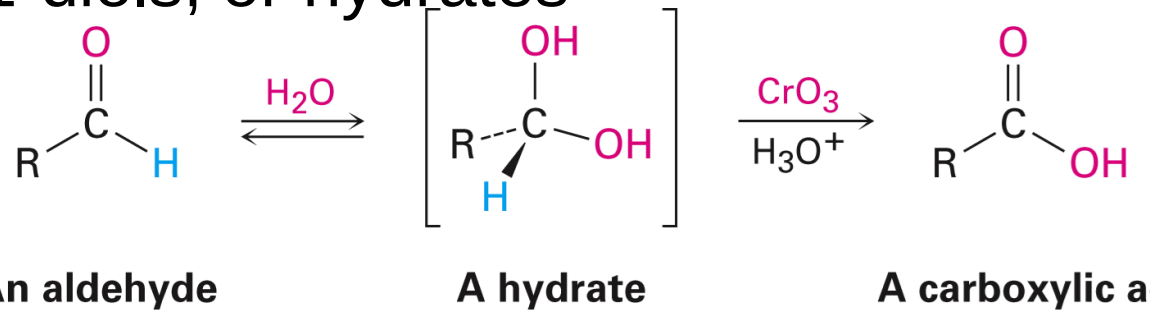
1-Hexyne **2-Hexanone (78%)**

Oxidation of Aldehydes and Ketones

- Aldehydes oxidize to yield carboxylic acids
  - CrO_3 in aqueous acid oxidizes aldehydes to carboxylic acids efficiently

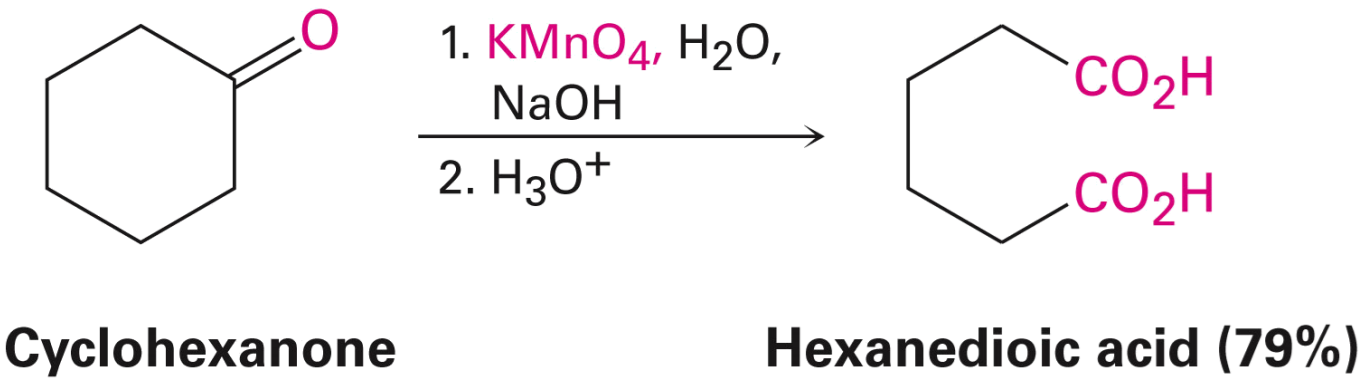


  - Aldehyde oxidations occur through intermediate 1,1-diols, or hydrates



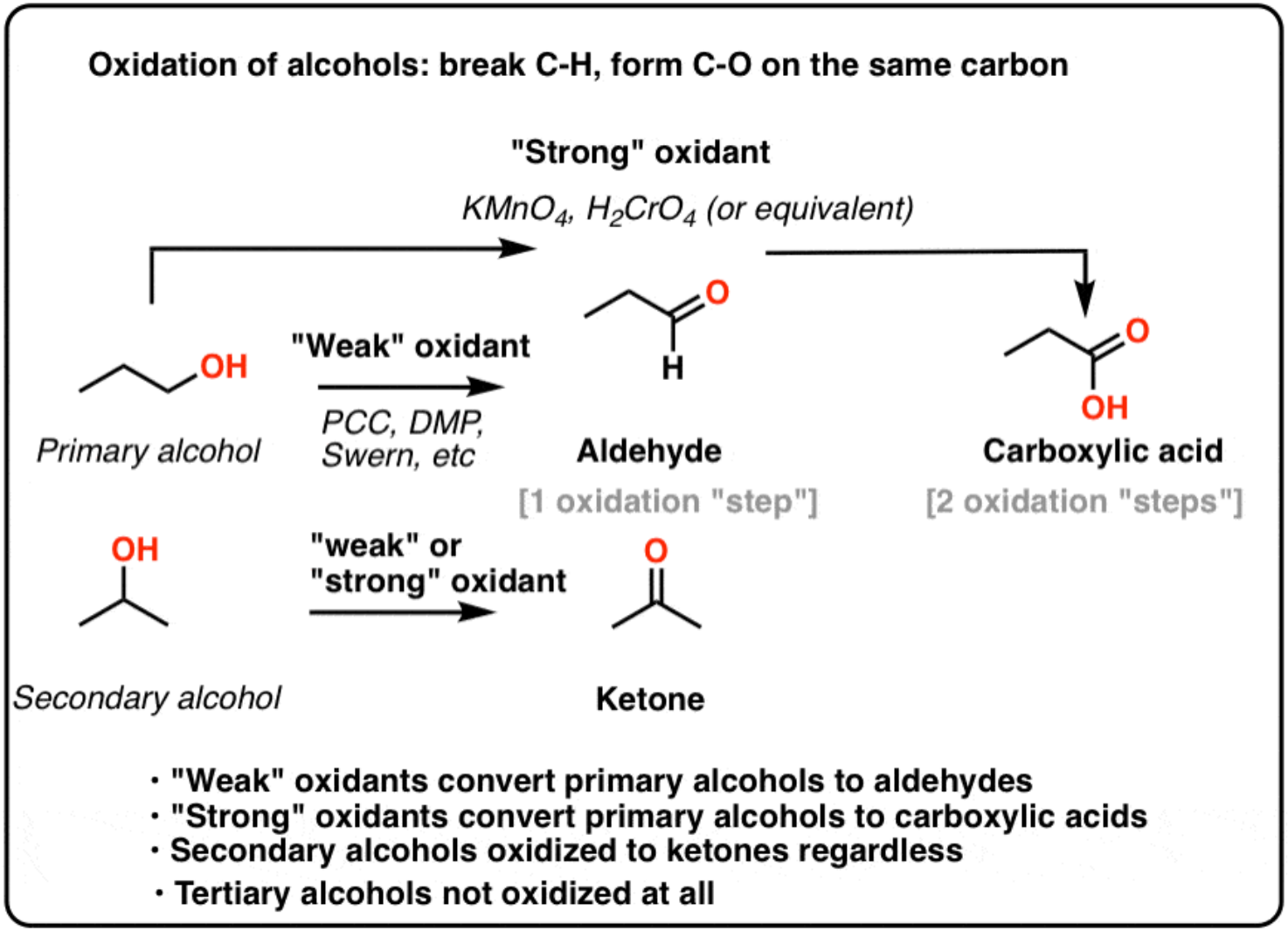
Oxidation of Aldehydes and Ketones

- Undergo slow cleavage with hot, alkaline KMnO4
- C–C bond next to C=O is broken to give carboxylic acids



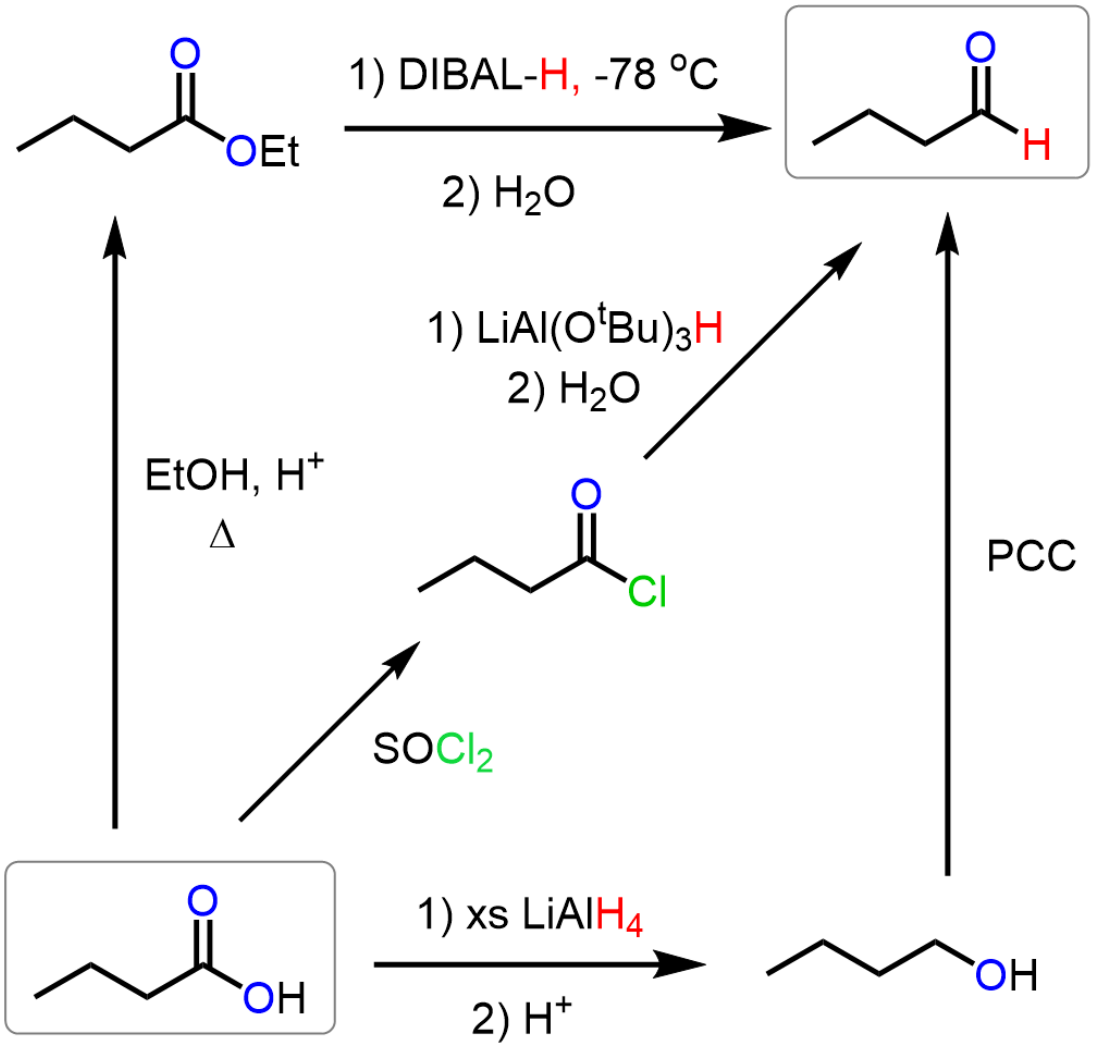
Cyclohexanone **Hexanedioic acid (79%)**

Alcohol Oxidation



https://www.masterorganicchemistry.com/

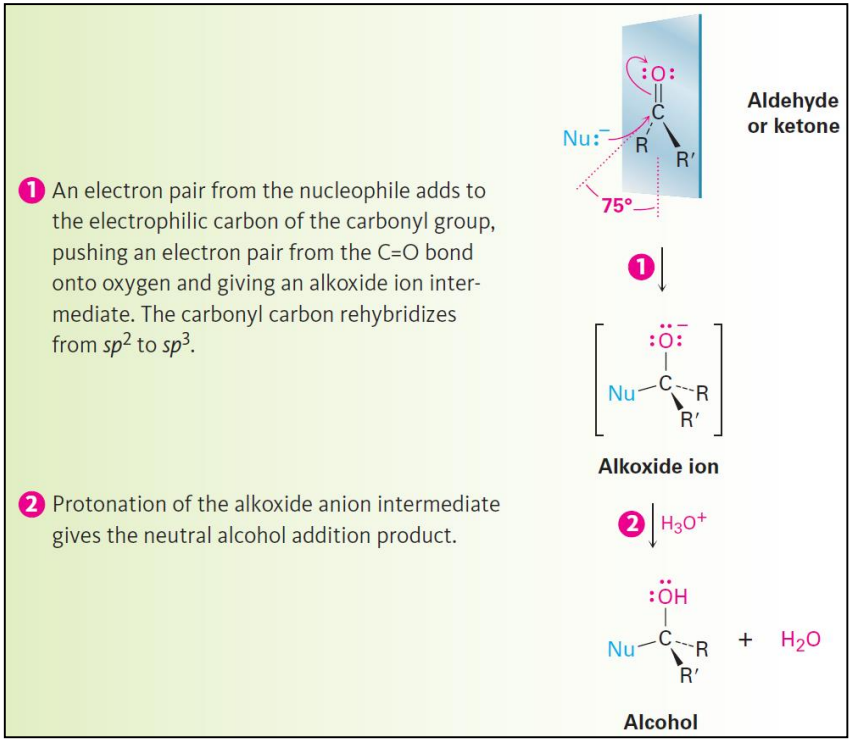
Reducing Carboxylic Acids to Aldehydes

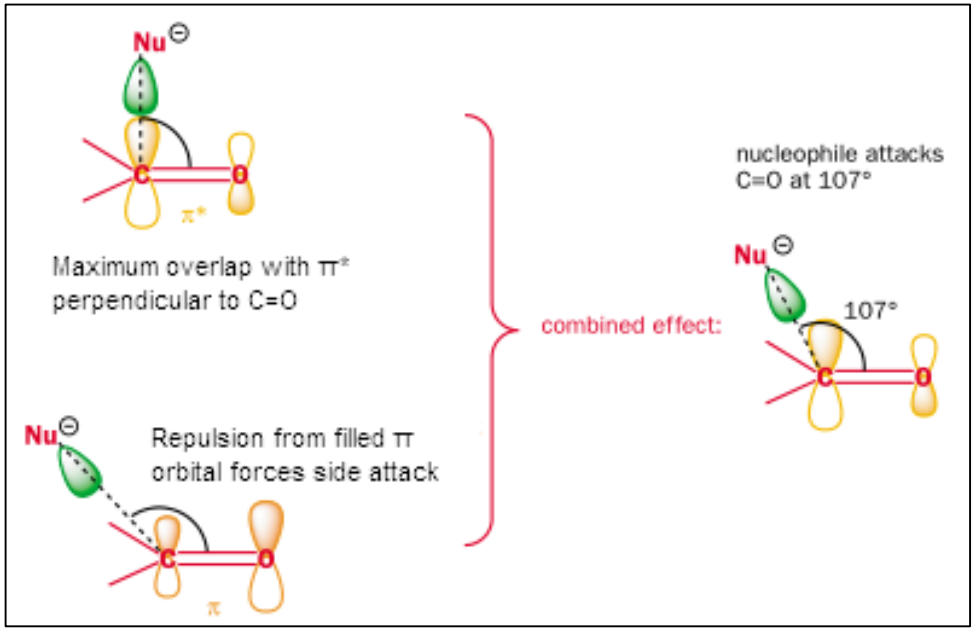


https://www.chemistrysteps.com/

Nucleophilic Addition Reactions of Aldehydes and Ketones

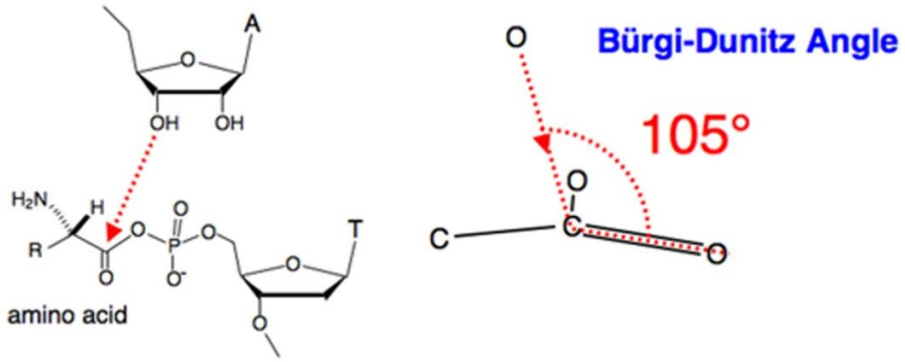
- Nu- approaches 75˚ to the plane of C=O and adds to C
- A tetrahedral alkoxide ion intermediate is produced





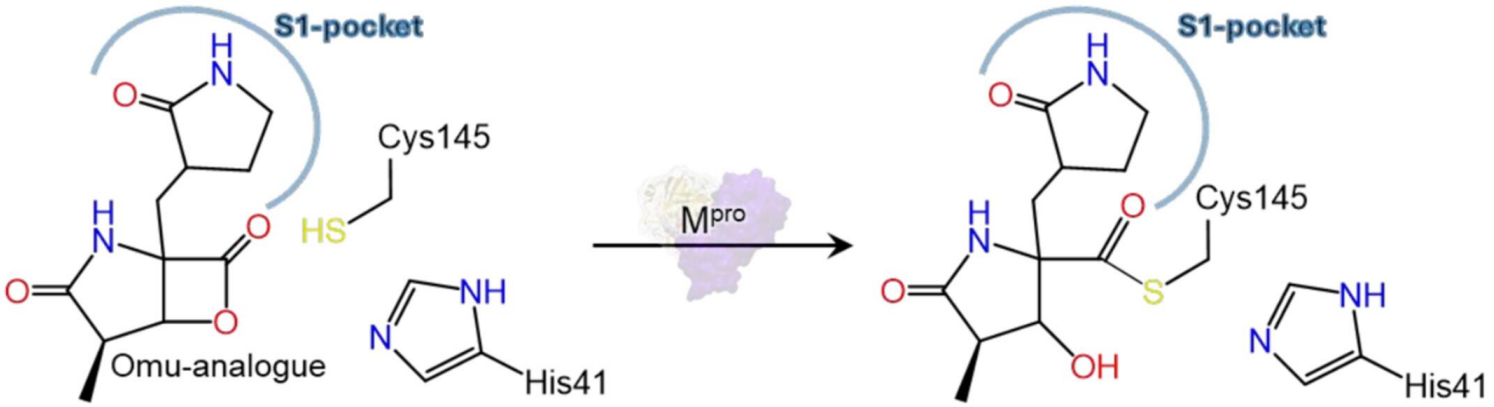
https://openchemistryhelp.blogspot.com/

Bürgi–Dunitz Geometry in Biological Reactions



Nucleophilic Attack by Ribose OH

Int. J. Mol. Sci. **2011**, 12



Enzyme:inhibitor complex **Enzyme-inhibitor covalent adduct**

Protein Science. **2026**, 35:e70680.

Nucleophilic Addition Reactions of Aldehydes and Ketones

- Nucleophiles can be negatively charged ($:Nu^-$) or neutral ($:Nu$) at the reaction site

Some negatively charged nucleophiles

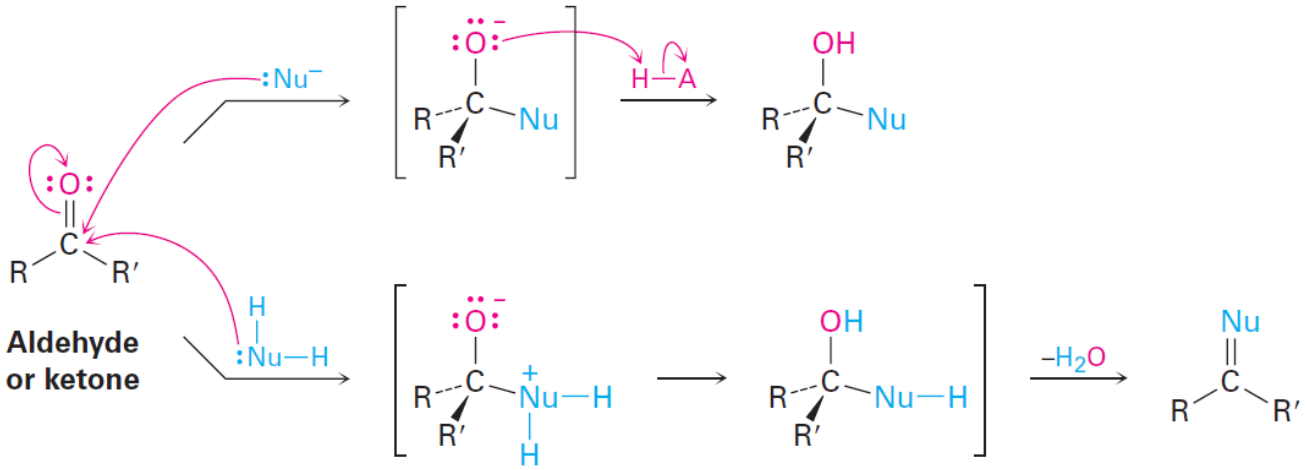
- $HO:^-$ (hydroxide ion)
- $H:^-$ (hydride ion)
- $R_3C:^-$ (a carbanion)
- $RO:^-$ (an alkoxide ion)
- $N{\equiv}C:^-$ (cyanide ion)

Some neutral nucleophiles

- HOH (water)
- ROH (an alcohol)
- $H_3N:$ (ammonia)
- RNH_2 (an amine)

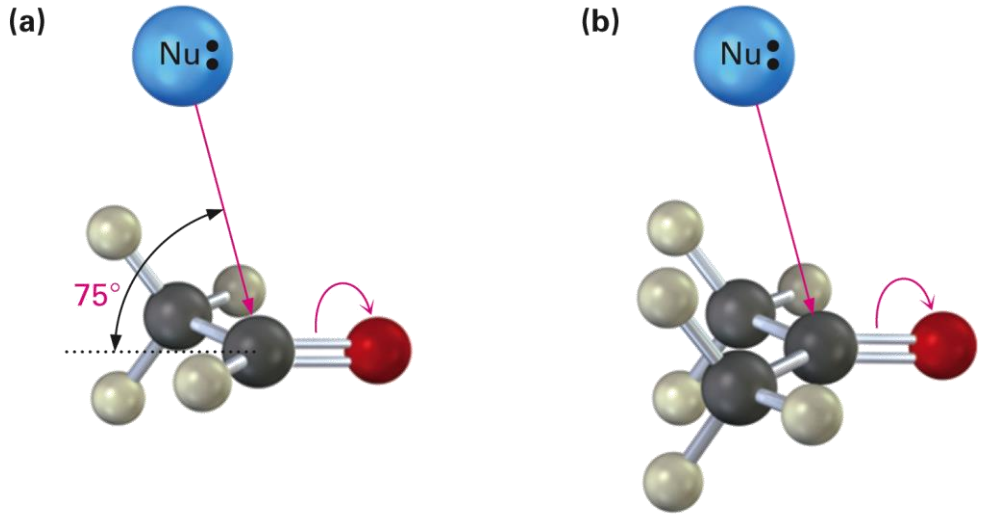
Nucleophilic Addition Reactions of Aldehydes and Ketones

- Nucleophilic additions to aldehydes and ketones have two general variations
  - Product is a direct result of the tetrahedral intermediate being protonated by water or acid
  - Carbonyl oxygen atom is protonated and eliminated as HO^- or H_2O to give a product with a C=Nu double bond



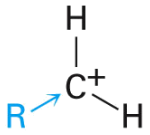
Nucleophilic Addition Reactions of Aldehydes and Ketones

- Aldehydes are reactive when compared to ketones in nucleophilic addition reactions
- Aldehydes have one large substituent bonded to the C=O, ketones have two
- The transition state for addition is less crowded and lower in energy for an aldehyde than for a ketone

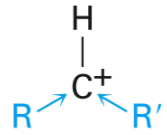


Electrophilicity of Aldehydes and Ketones

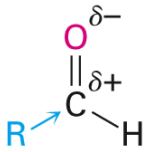
- Aldehydes are more polarized than ketones
- In carbocations, more alkyl groups stabilize the positive charge
- Ketone has more alkyl groups, stabilizing the C=O carbon inductively



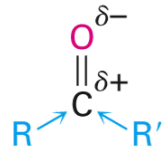
1° carbocation
(less stable, more reactive)



2° carbocation
(more stable, less reactive)



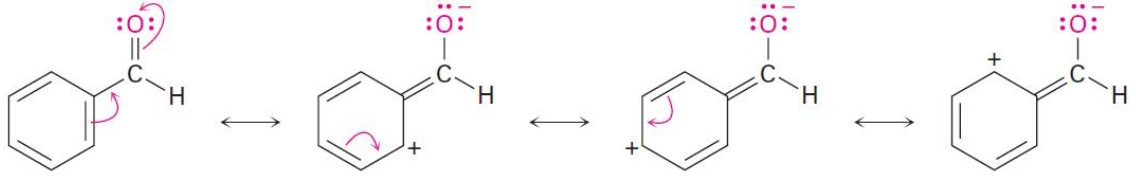
Aldehyde
(less stabilization of δ+, more reactive)

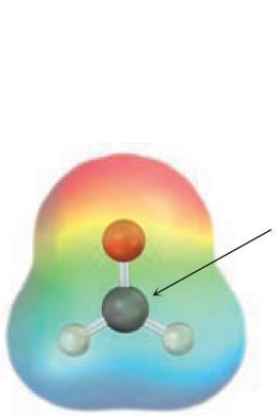


Ketone
(more stabilization of δ+, less reactive)

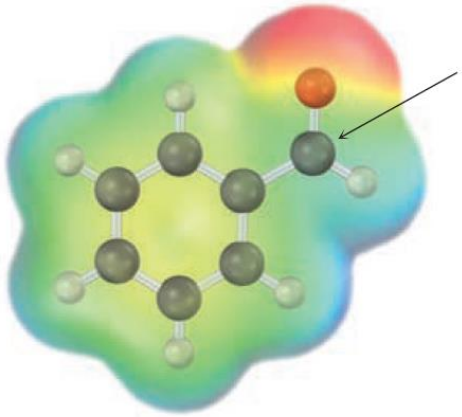
Reactivity of Aromatic Aldehydes

- Less reactive in nucleophilic addition reactions than aliphatic aldehydes
- Example - Carbonyl carbon atom is less positive in the aromatic aldehyde





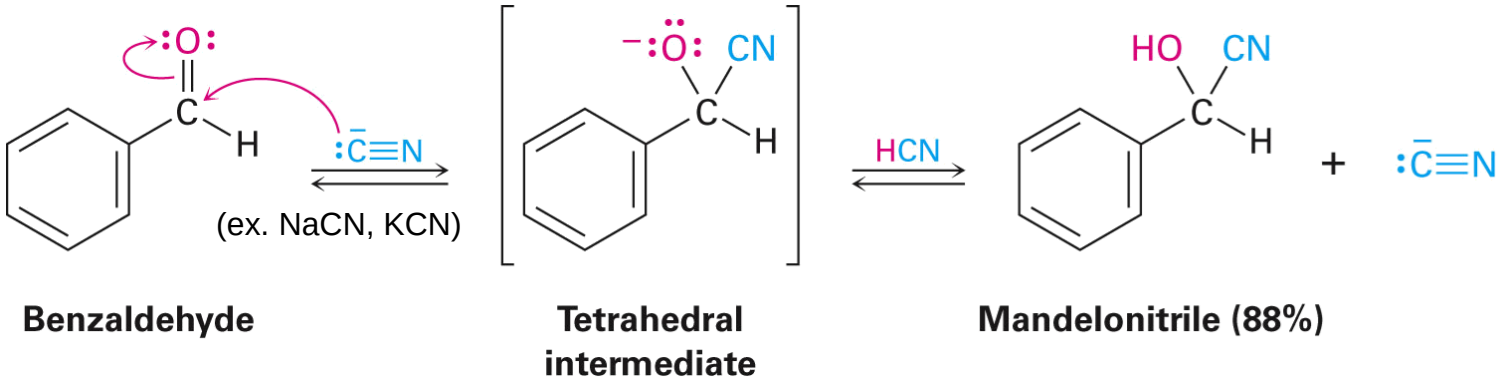
Formaldehyde



Benzaldehyde

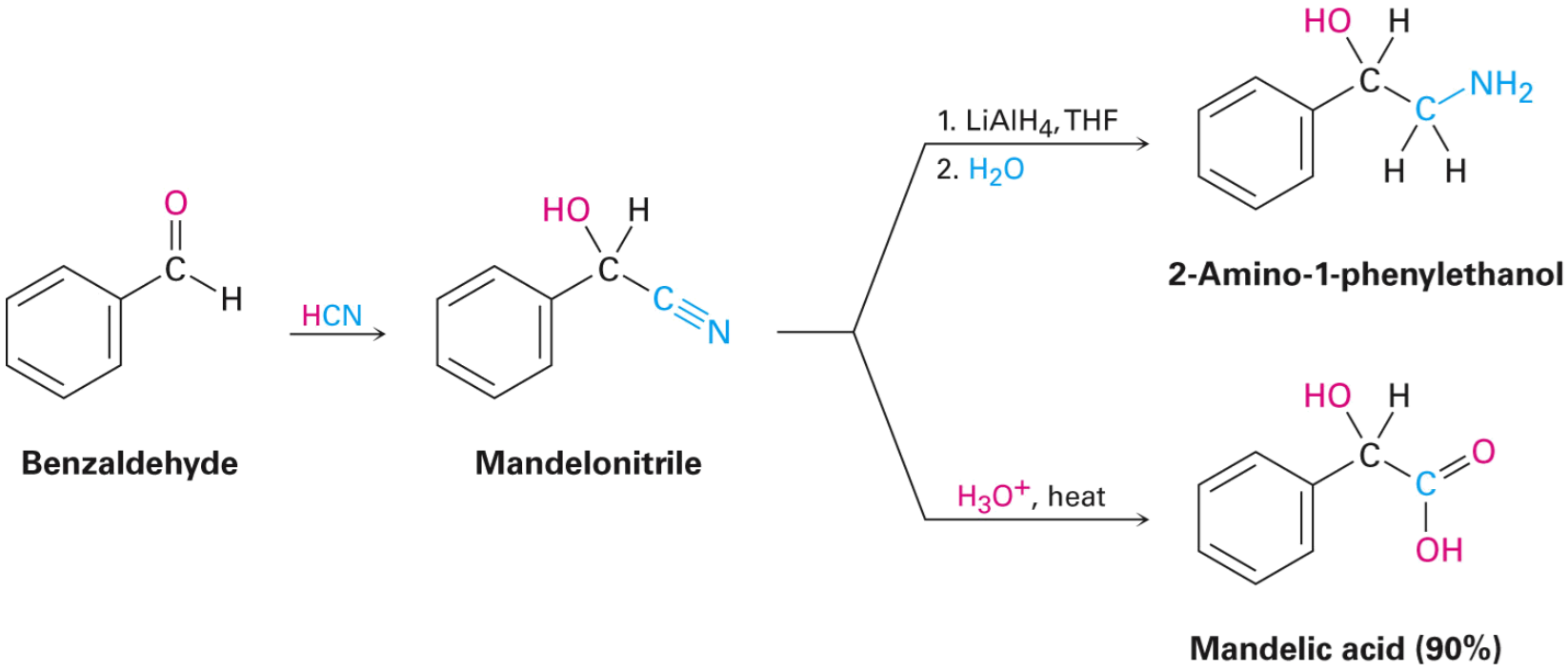
Nucleophilic Addition of HCN: Cyanohydrin Formation

- **Cyanohydrins**: Product of nucleophilic reaction between aldehydes and unhindered ketones with HCN
  - Addition of HCN is reversible and base-catalyzed, generating nucleophilic cyanide ion, CN^-
  - Addition of CN^- to C=O yields a tetrahedral intermediate, which is then protonated
  - Equilibrium favors cyanohydrin adduct



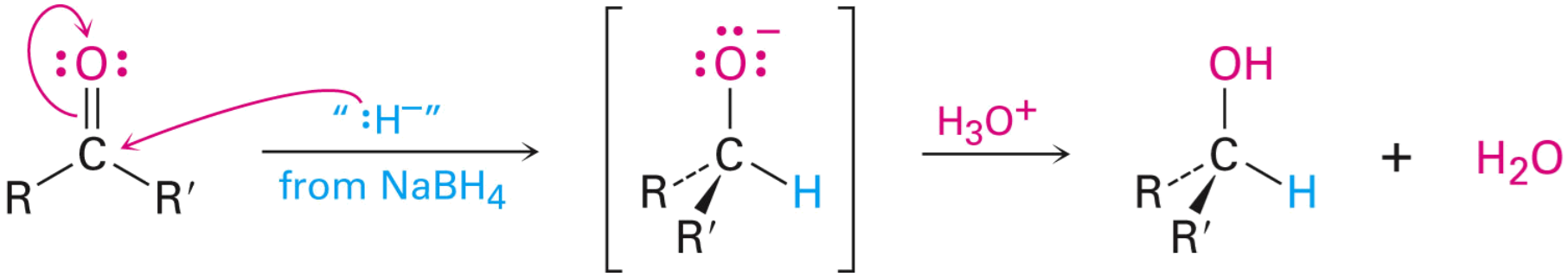
Uses of Cyanohydrins

- The nitrile group (R–C≡N) can be reduced with $LiAlH_4$ to yield a primary amine (RCH_2NH_2)
- Can be hydrolyzed by hot acid to yield a carboxylic acid



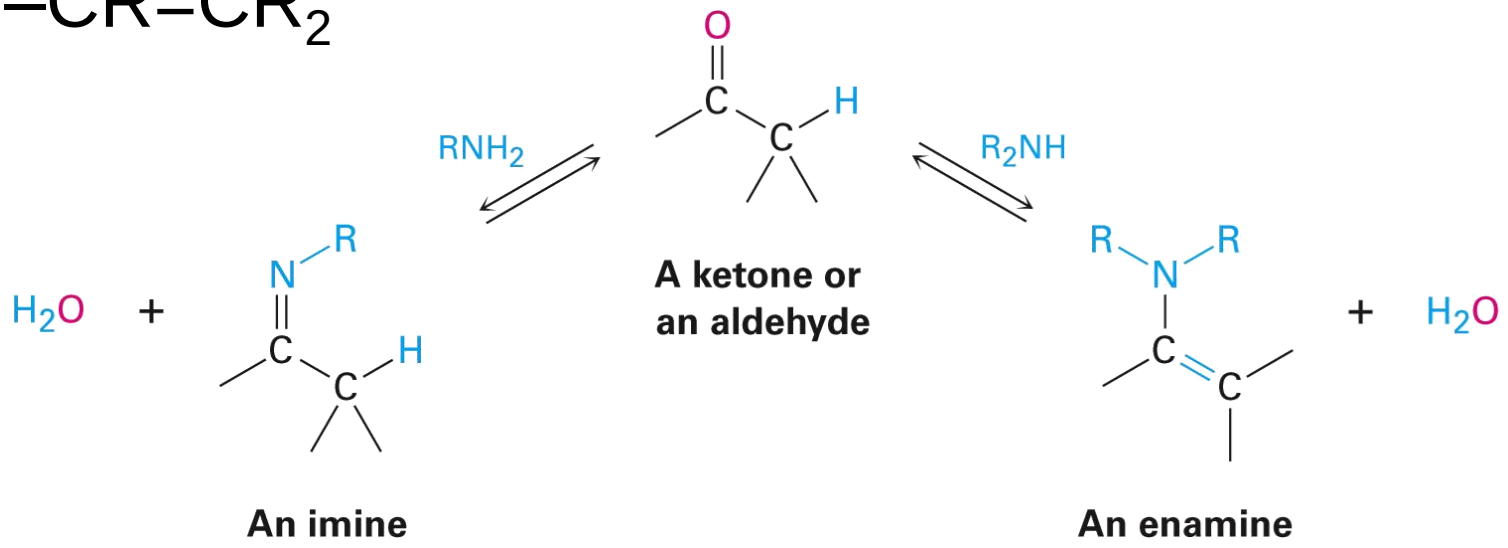
Nucleophilic Addition of Grignard Reagents and Hydride Reagents: Alcohol Formation

- $LiAlH_4$ and $NaBH_4$ react as donors of hydride ion
- Protonation after addition yields the alcohol
- Reaction is effectively irreversible



Nucleophilic Addition of Amines: Imine and Enamine Formation

- RNH_2 adds to aldehydes and keytones to form **imines**, $R_2C{=}NR$
- R_2NH adds similarly to yield **enamines**,
- $R_2N{-}CR{=}CR_2$



- Imines are common as intermediates in biological pathways, and are called **Schiff bases**

Figure 19.6 - Mechanism

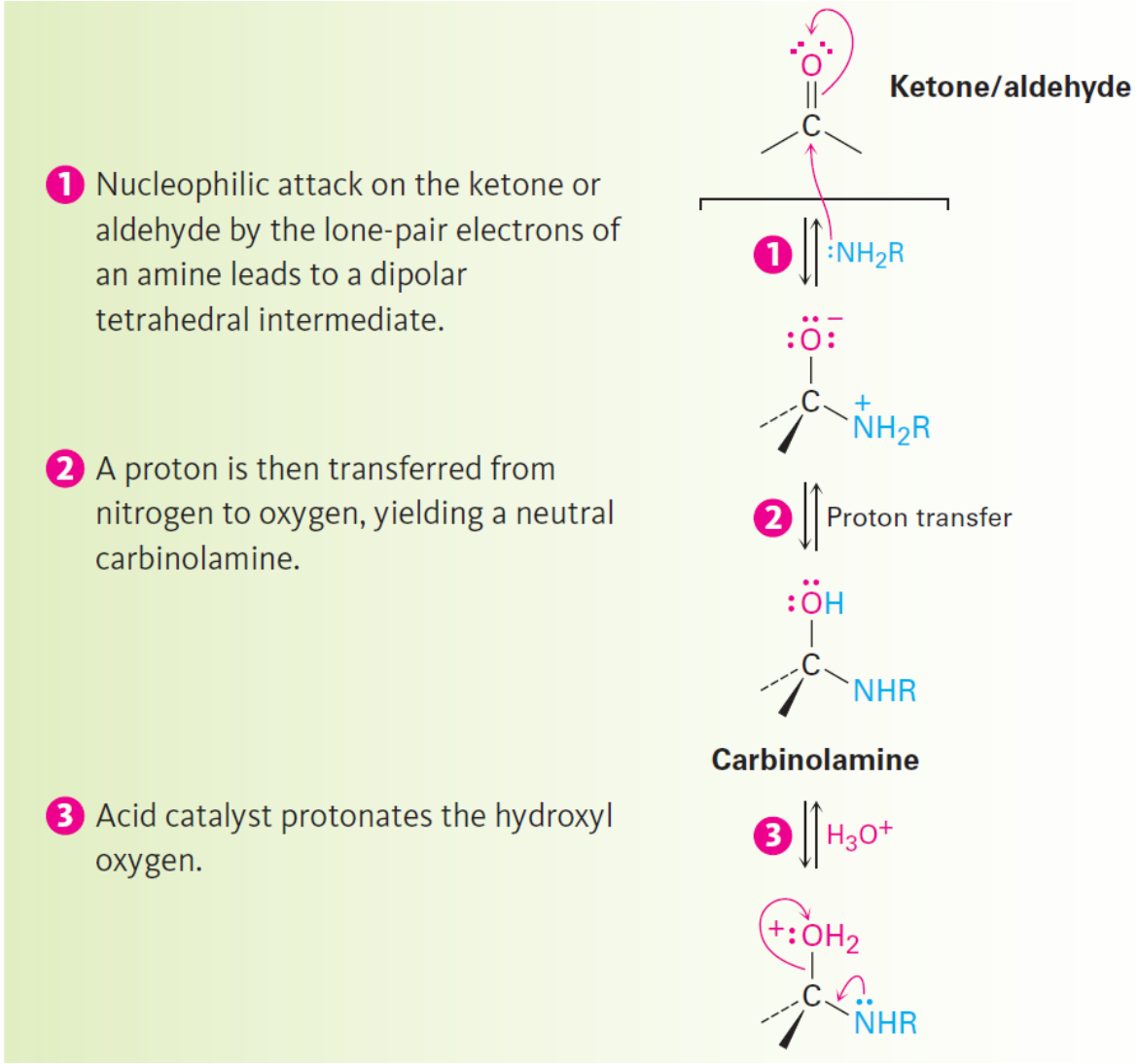


Figure 19.6 - Mechanism

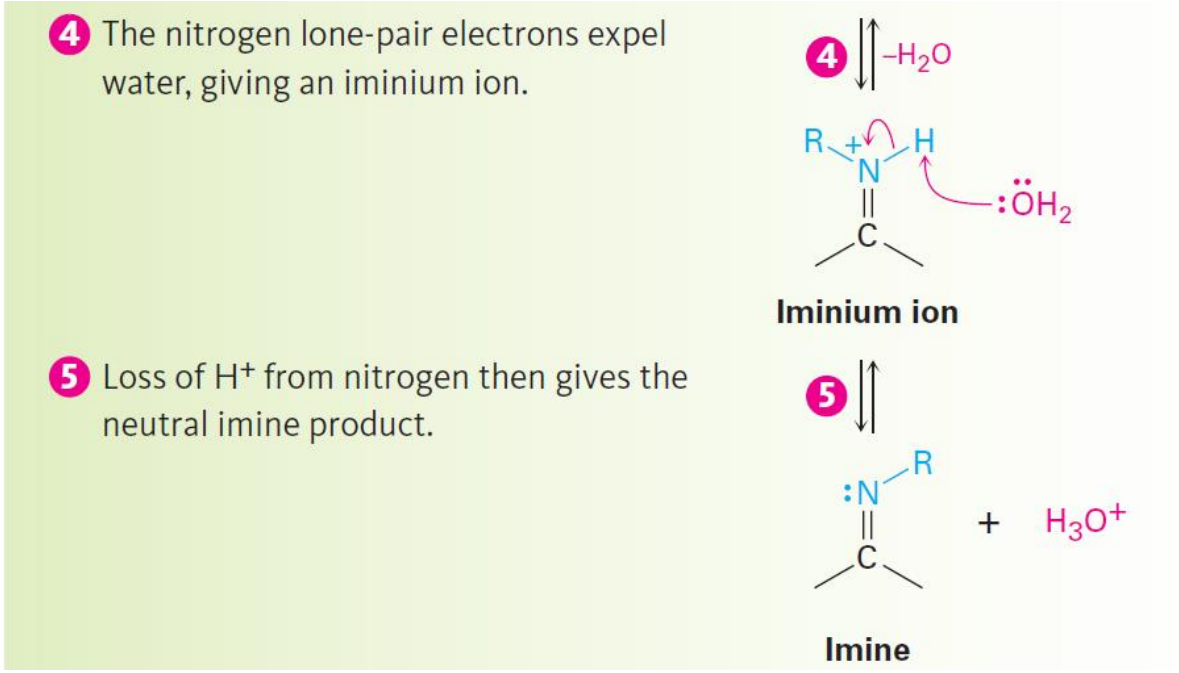
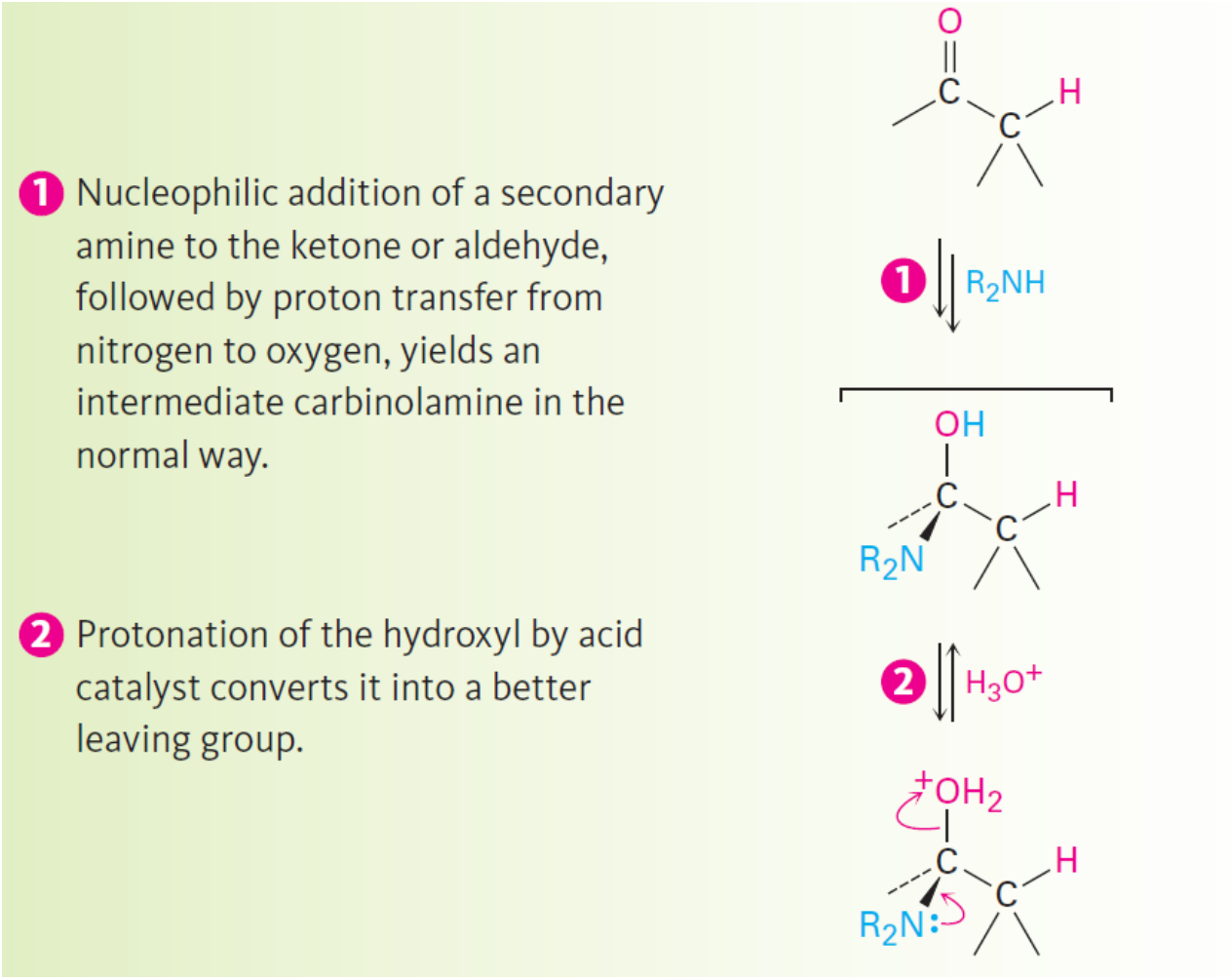


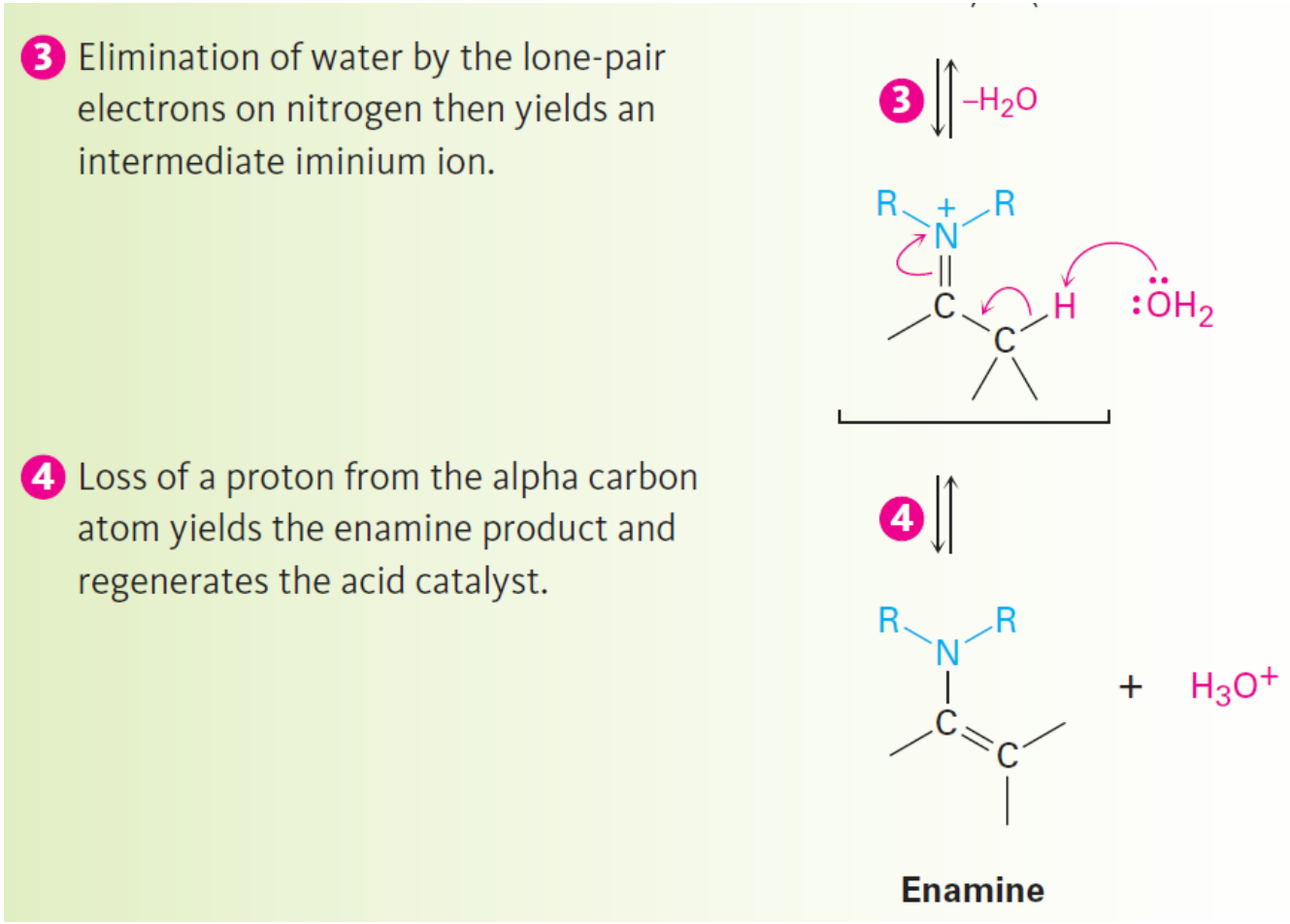
Figure 19.7 - Enamine Formation



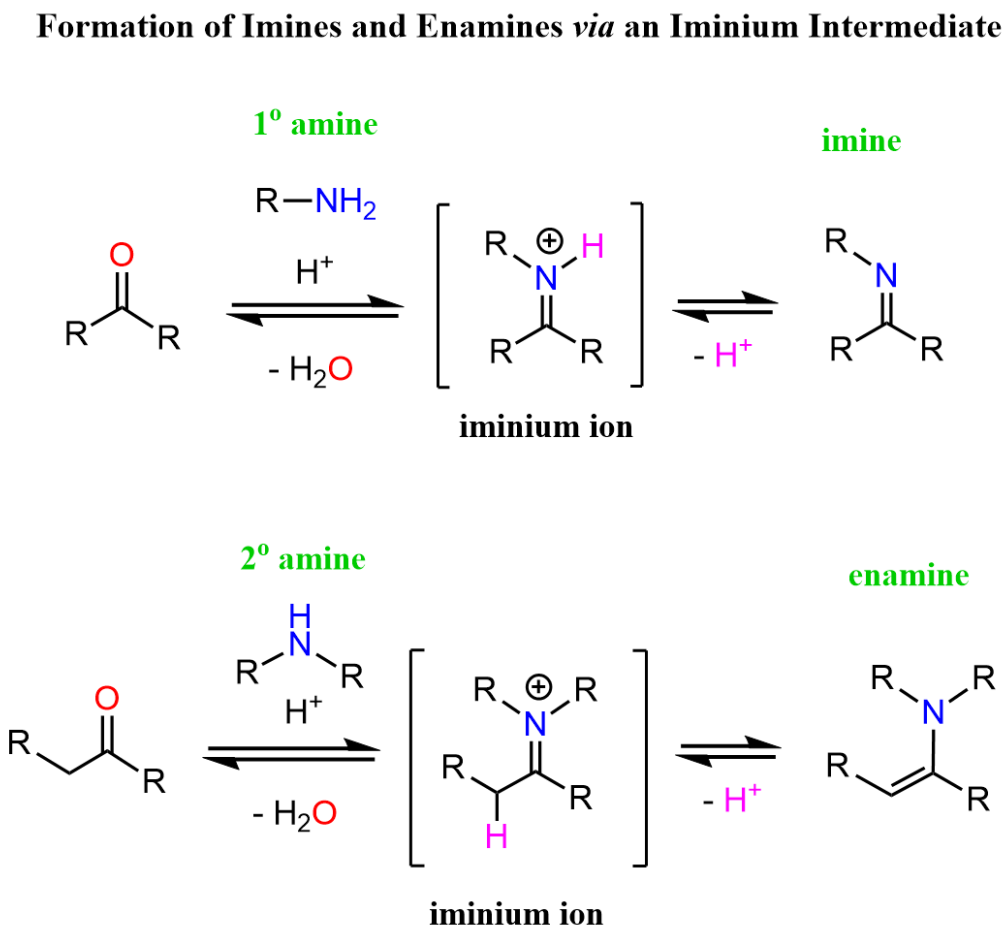
1. Nucleophilic addition of a secondary amine to the ketone or aldehyde, followed by proton transfer from nitrogen to oxygen, yields an intermediate carbinolamine in the normal way.

2. Protonation of the hydroxyl by acid catalyst converts it into a better leaving group.

Figure 19.7 - Enamine Formation

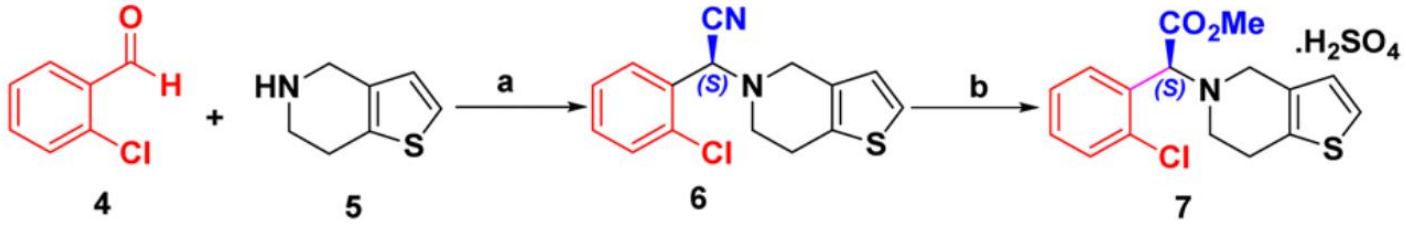


Example



https://www.chemistrysteps.com/

The Strecker reaction generates an iminium salt intermediate





antiplatelet drug

Reaction conditions (a) **4** (3 g, 21 mmol), **5** (2.9 g, 21 mmol), TMSCN (2.0 g, 21 mmol), catalyst V (0.2 mol%), DCM (20 mL) to give **6** (98% yield, 99.3% ee); (b) MeOH, H_2SO_4 to give **7** (73.2% yield, 98.5% ee)

Clopidogrel (Plavix)
Sanofi (1997 FDA)

Chem. Commun., **2024**, 60, 8561–8564

Nucleophilic Addition of Hydrazine: The Wolff-Kishner Reaction

- Treatment of an aldehyde or ketone with hydrazine, H_2NNH_2, and KOH to convert the compound to an alkane
- Involves formation of a hydrazone intermediate, $R_2C{=}NNH_2$, followed by:
  - Base-catalyzed double-bond migration
  - Loss of N_2 gas to give a carbanion
  - Protonation to give the alkane product
- More useful than catalytic hydrogenation

Figure 19.9 - Mechanism

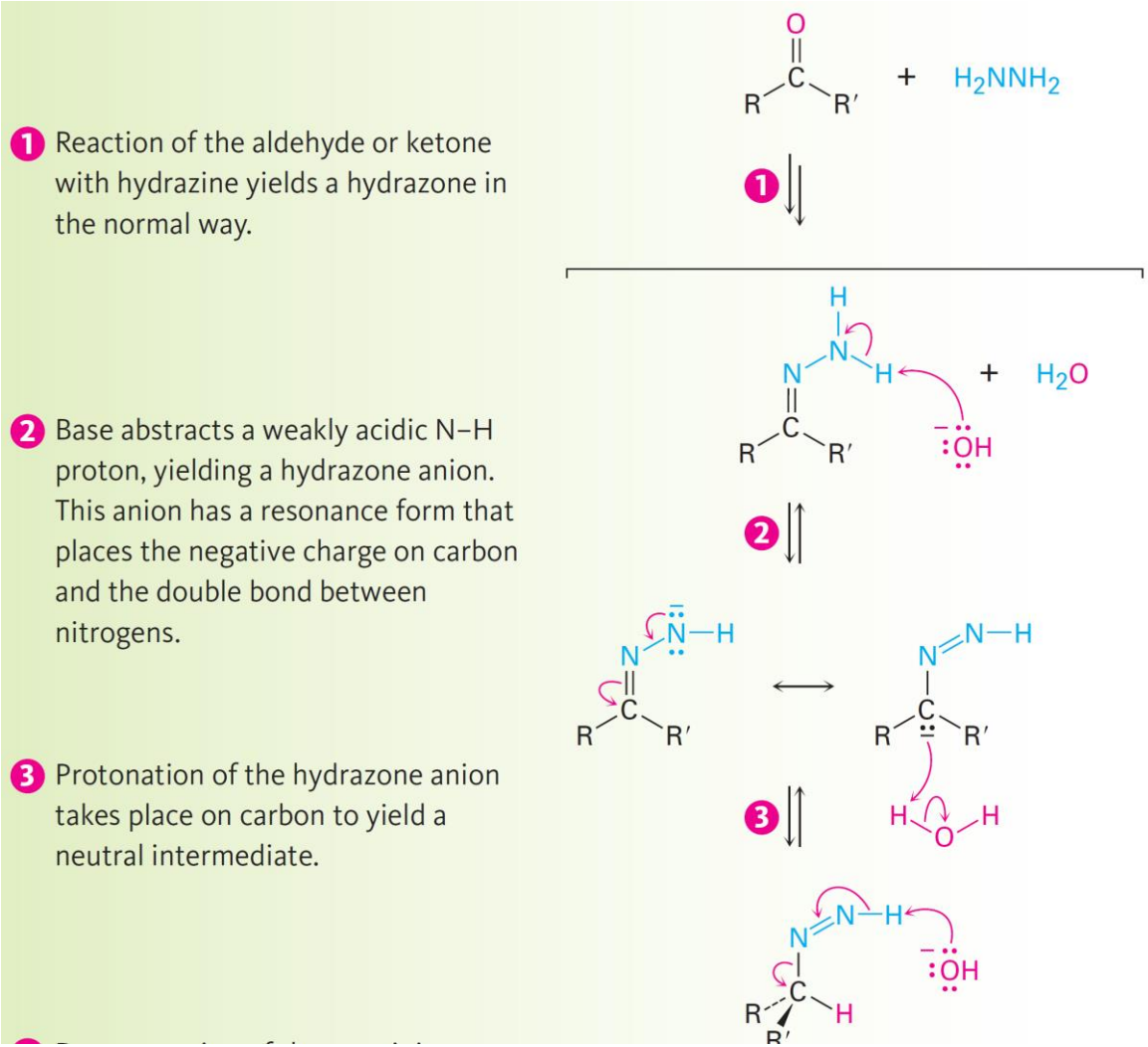
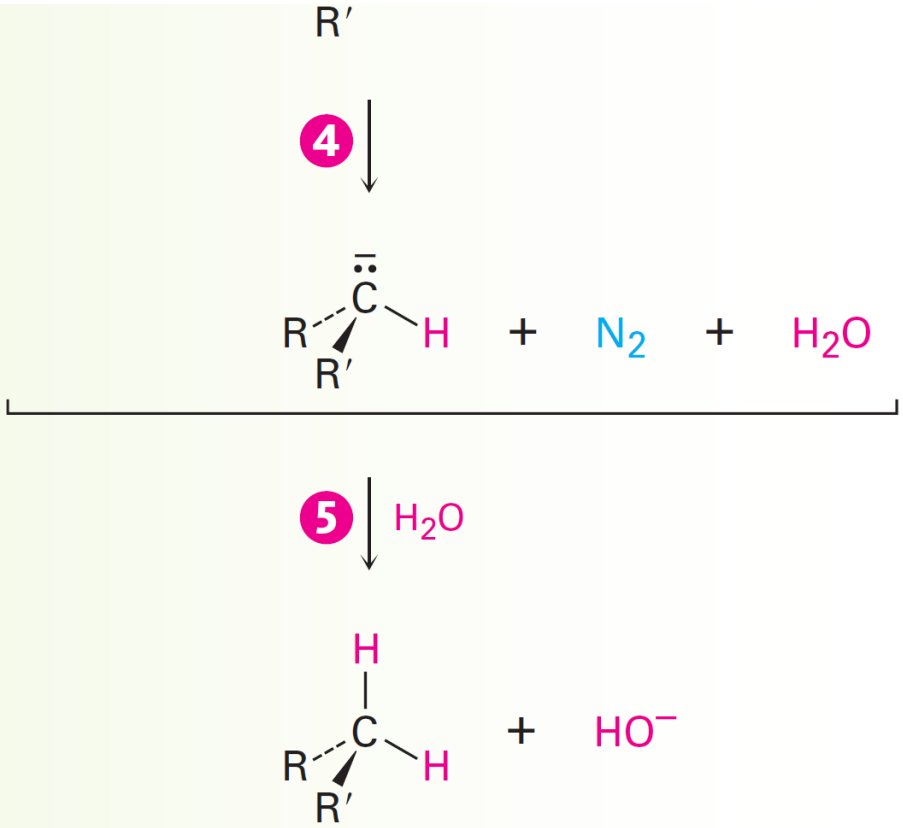


Figure 19.9 - Mechanism

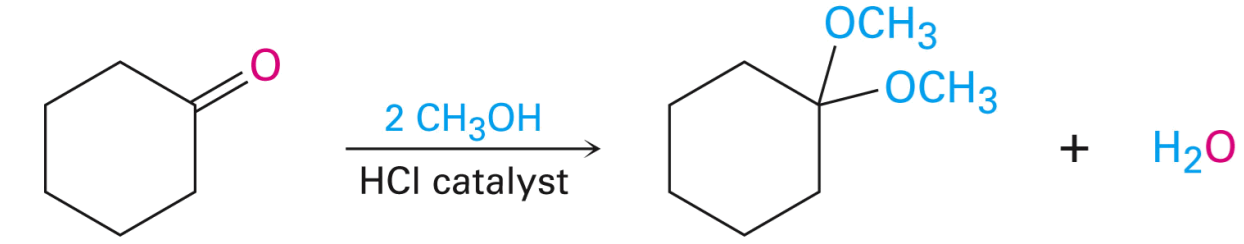
4 Deprotonation of the remaining weakly acidic N–H occurs with simultaneous loss of nitrogen to give a carbanion . . .

5 . . . which is protonated to give the alkane product.



Nucleophilic Addition of Alcohols: Acetal Formation

- Aldehydes and ketones react reversibly with 2 equivalents of an alcohol in the presence of an acid catalyst to yield **acetals, $R_2C(OR')_2$**
  - Called ketals if derived from a ketone



Cyclohexanone **Cyclohexanone dimethyl acetal**

- Under acidic conditions reactivity of the carbonyl group is increased by protonation, so addition of an alcohol occurs rapidly

Nucleophilic Addition of Alcohols: Acetal Formation

- Nucleophilic addition of an alcohol to the carbonyl group initially yields a hydroxy ether called a **hemiacetal**
  - Formed reversibly
- Reaction can be driven either forward or backward depending on the conditions

Figure 19.10 - Mechanism

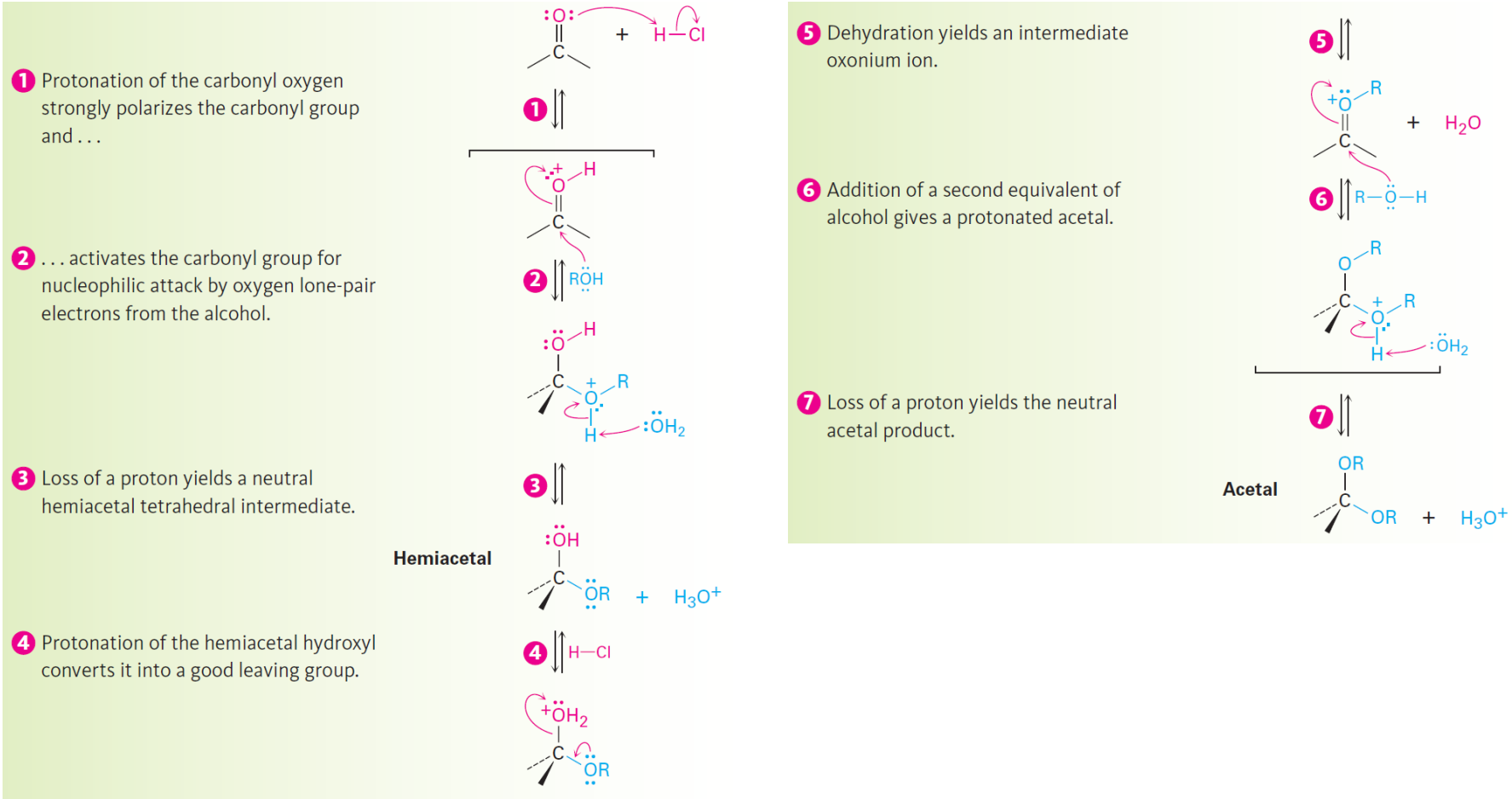
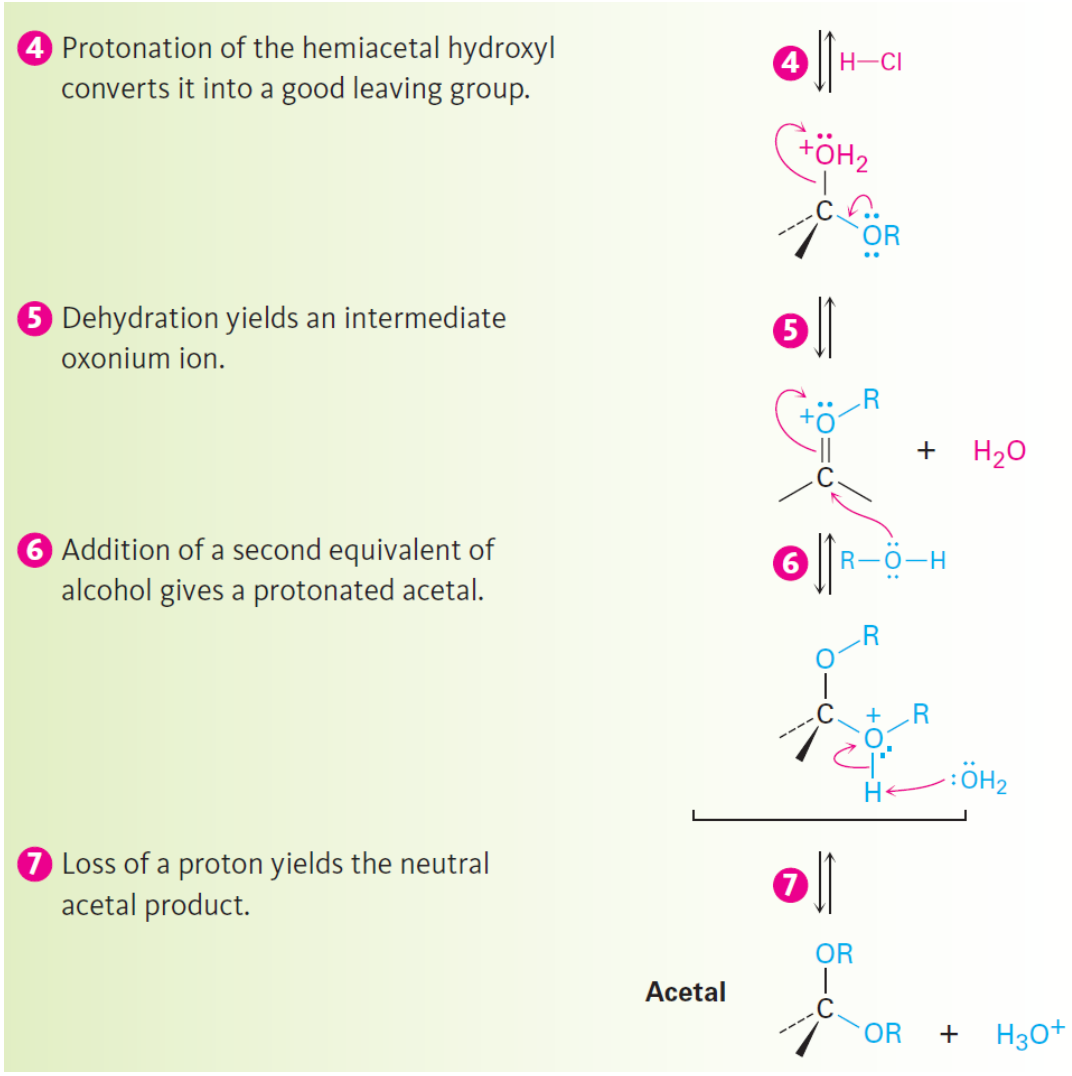
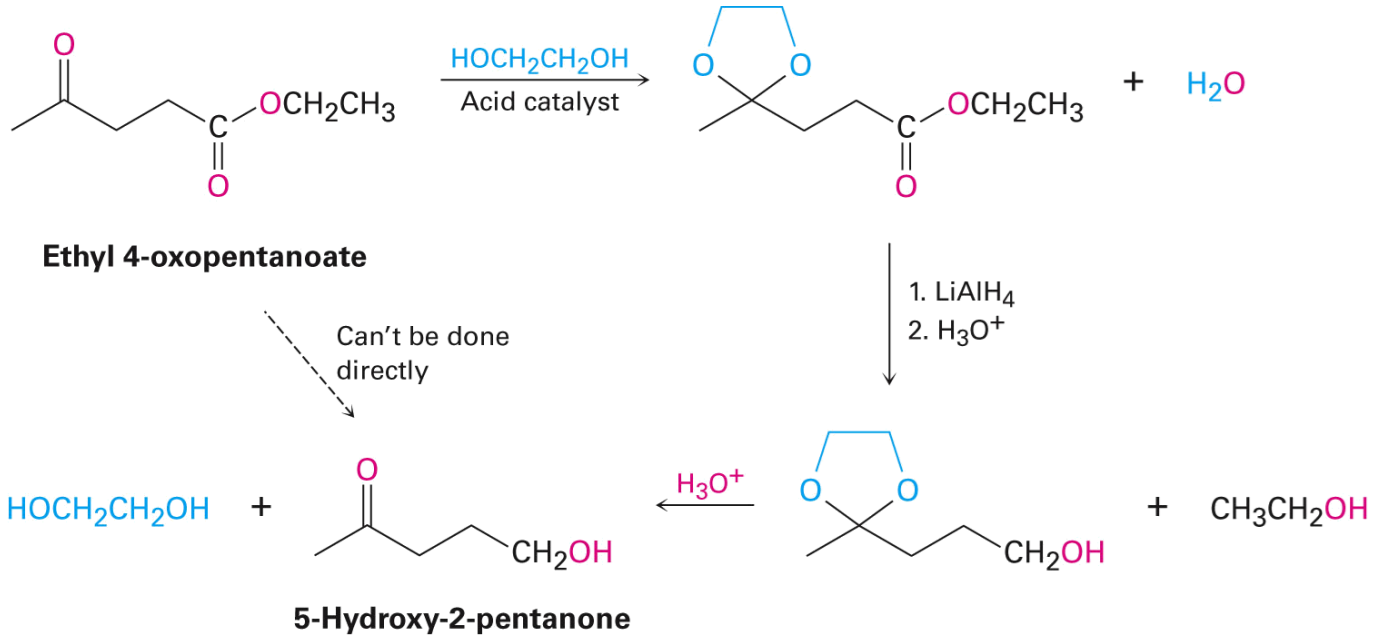


Figure 19.10 - Mechanism



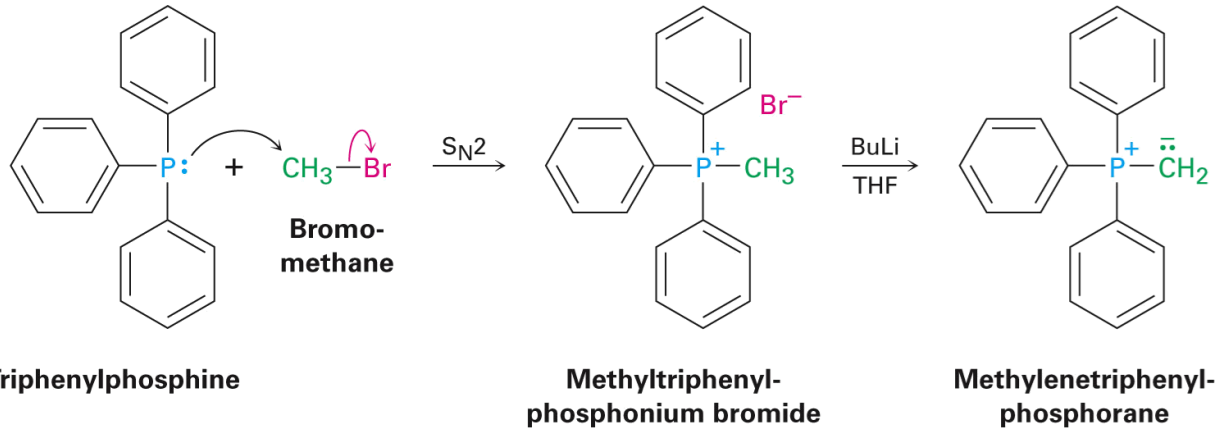
Uses of Acetals

- Acetals can serve as protecting groups for aldehydes and ketones
- Easier to use a diol to form a cyclic acetal



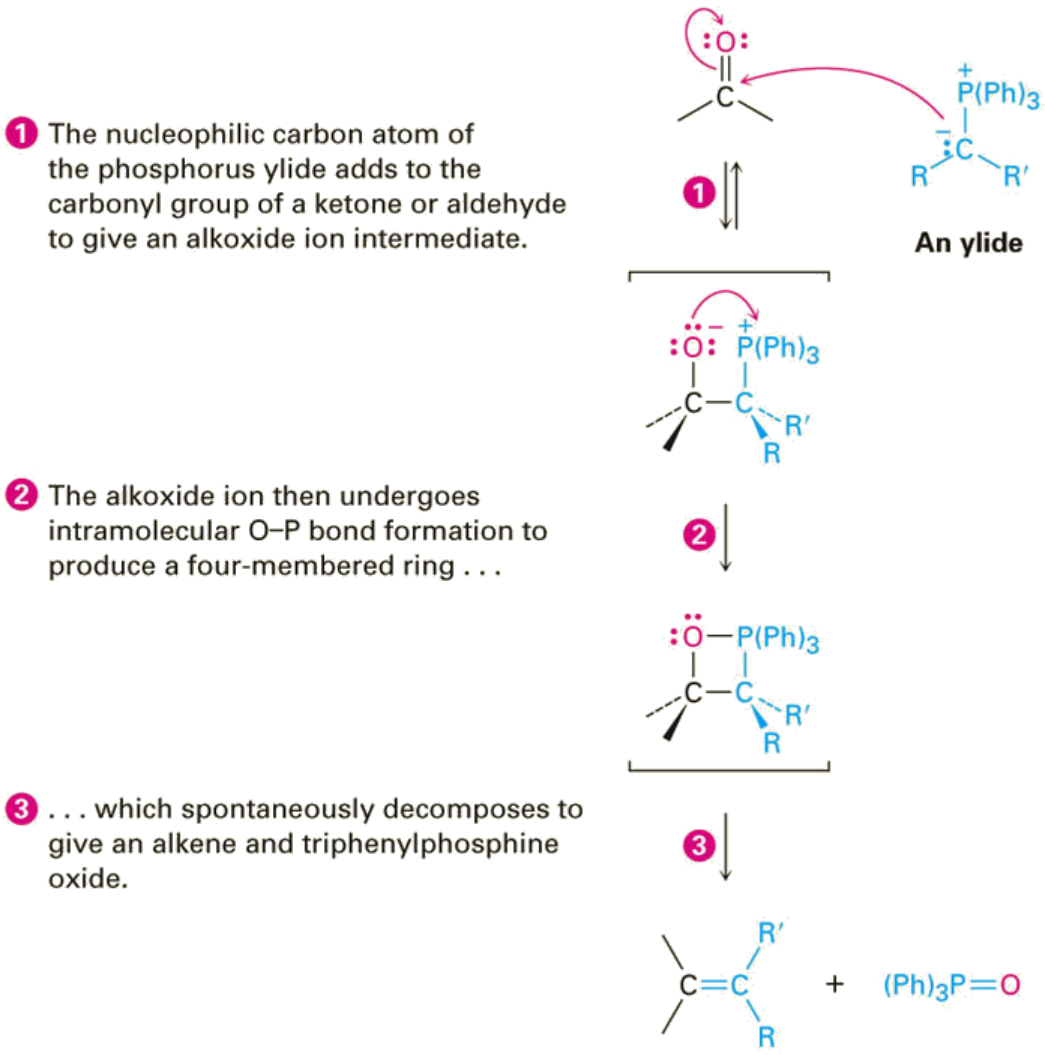
Nucleophilic Addition of Phosphorus Ylides: The Wittig Reaction

- Triphenylphosphine is a good nucleophile in S_N2 reactions
  - Yields alkyltriphenylphosphonium salts



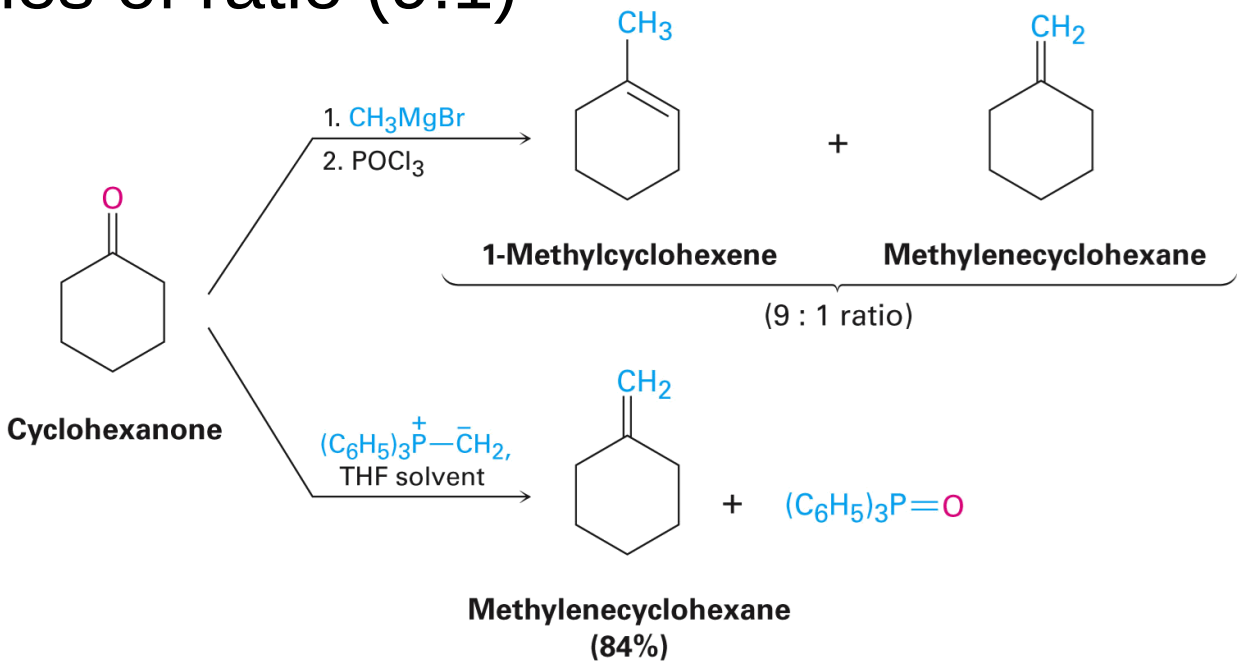
- Cannot be used to prepare tetrasubstituted alkenes due to steric hindrance

Mechanism of the Wittig Reaction



Nucleophilic Addition of Phosphorus Ylides: The Wittig Reaction

- Addition of CH_3MgBr to cyclohexanone and dehydration with $POCl_3$, yields a mixture of two alkenes of ratio (9:1)



Biological Reductions

- **Cannizzaro reaction**: Nucleophilic addition of OH^- to an aldehyde to give a tetrahedral intermediate, which expels hydride ion as a leaving group and is thereby oxidized
  - A second aldehyde molecule accepts the hydride ion in another nucleophilic addition step and is thereby reduced

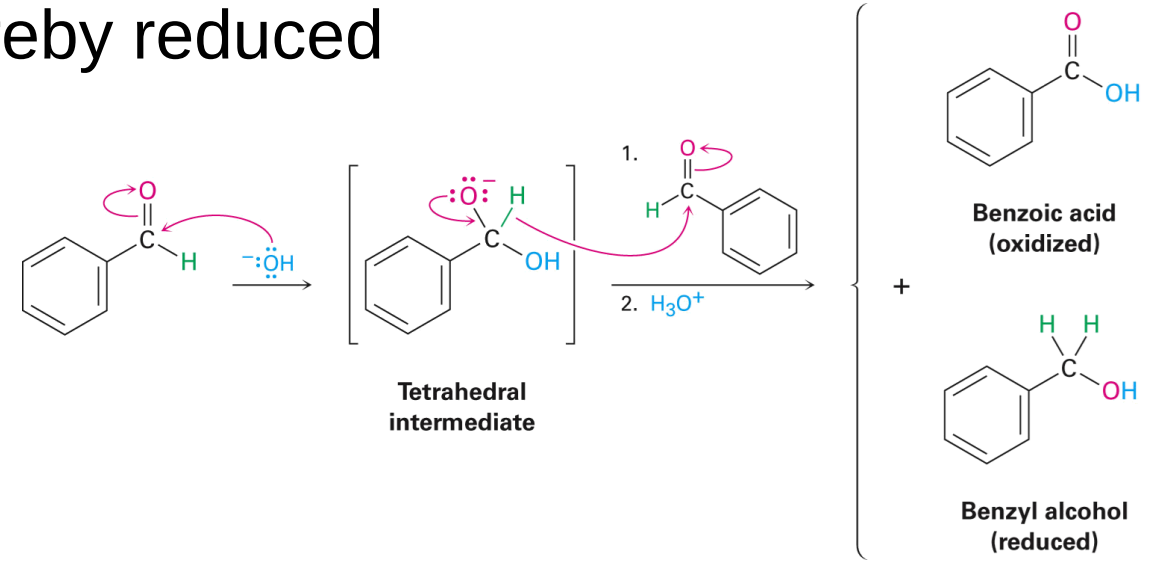
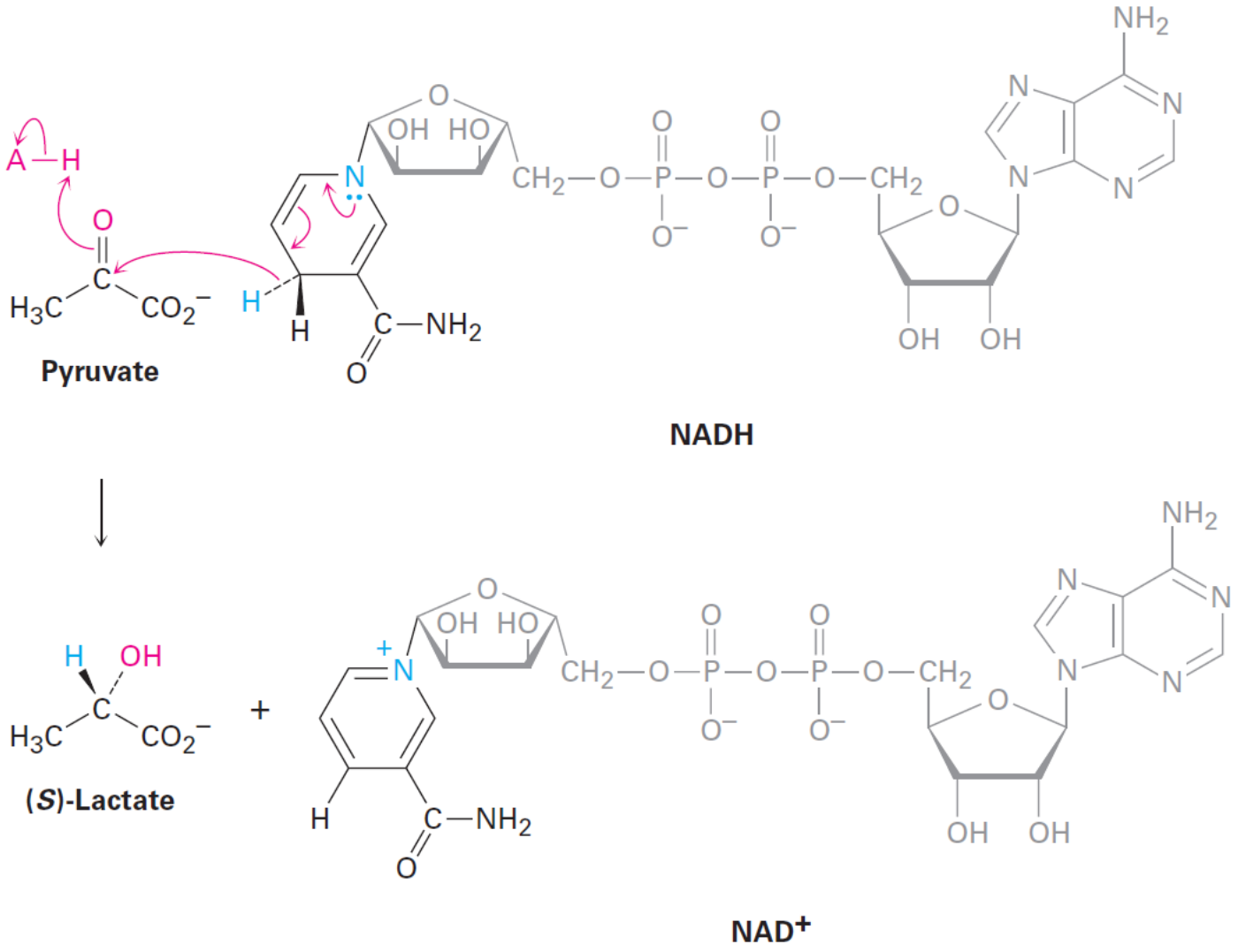
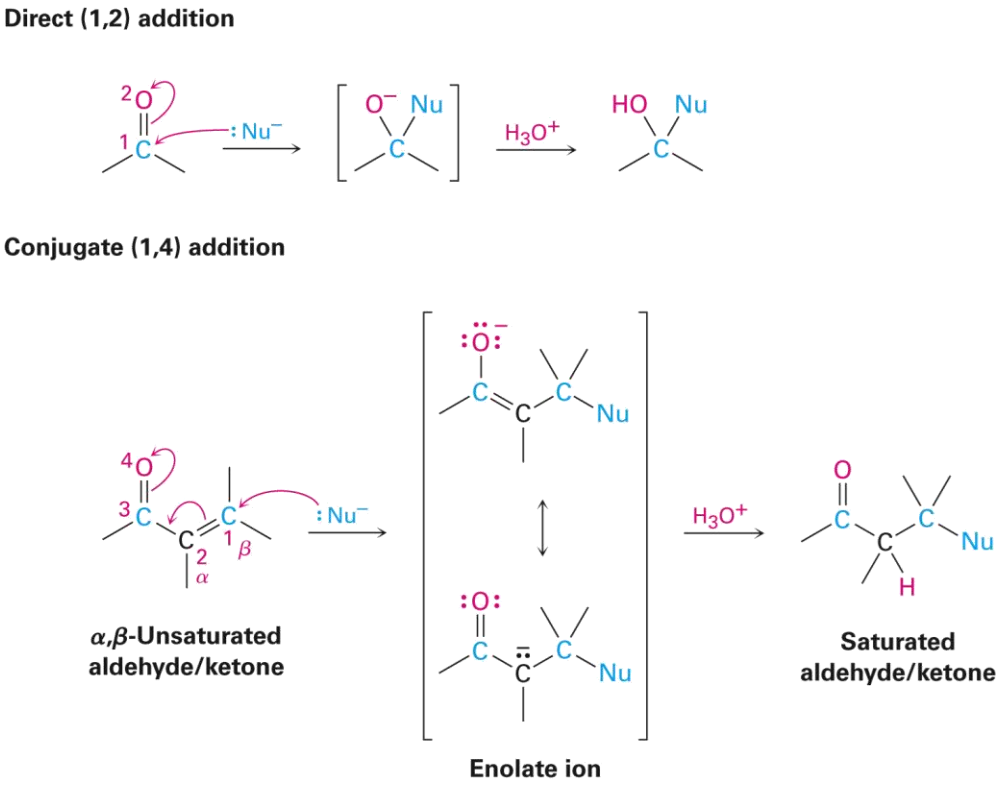


Figure 19.12 - Mechanism of Biological Aldehyde and Ketone Reductions



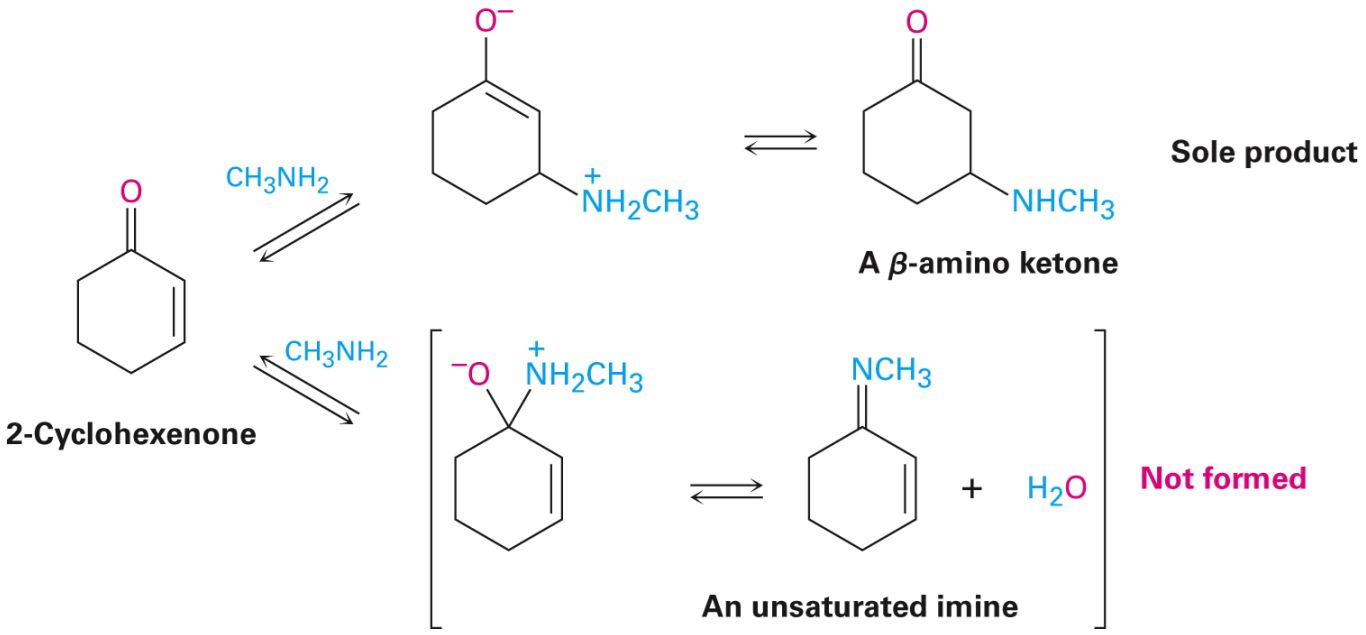
Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

- **1,2-addition**: Addition of a nucleophile directly to the carbonyl group
- **Conjugate addition (1,4-addition)**: Addition of a nucleophile to the C=C double bond of an α,β-unsaturated aldehyde or ketone



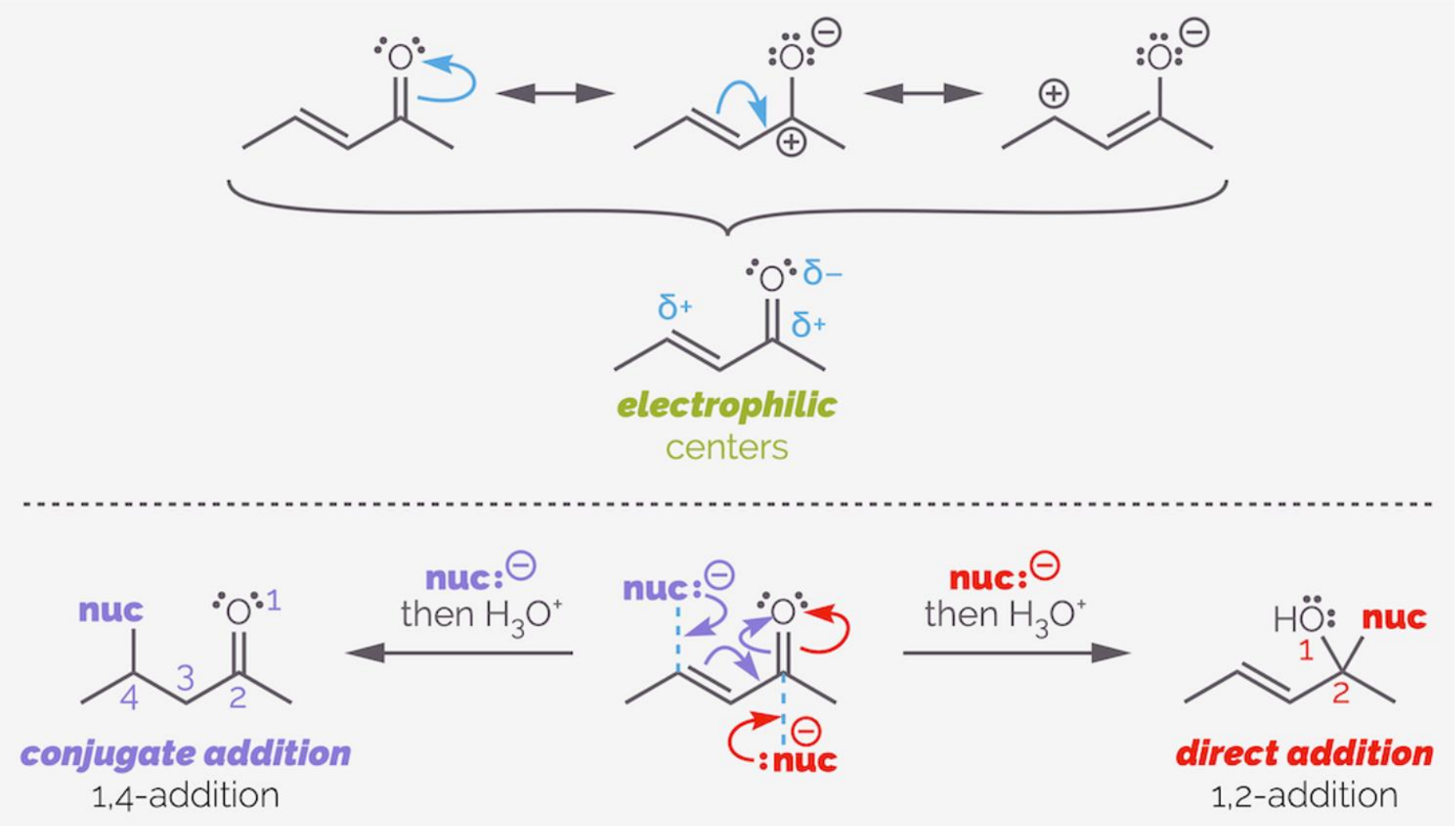
Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

- Conjugate addition of amines
  - Primary and secondary amines add to α, β - unsaturated aldehydes and ketones to yield β-amino aldehydes and ketones



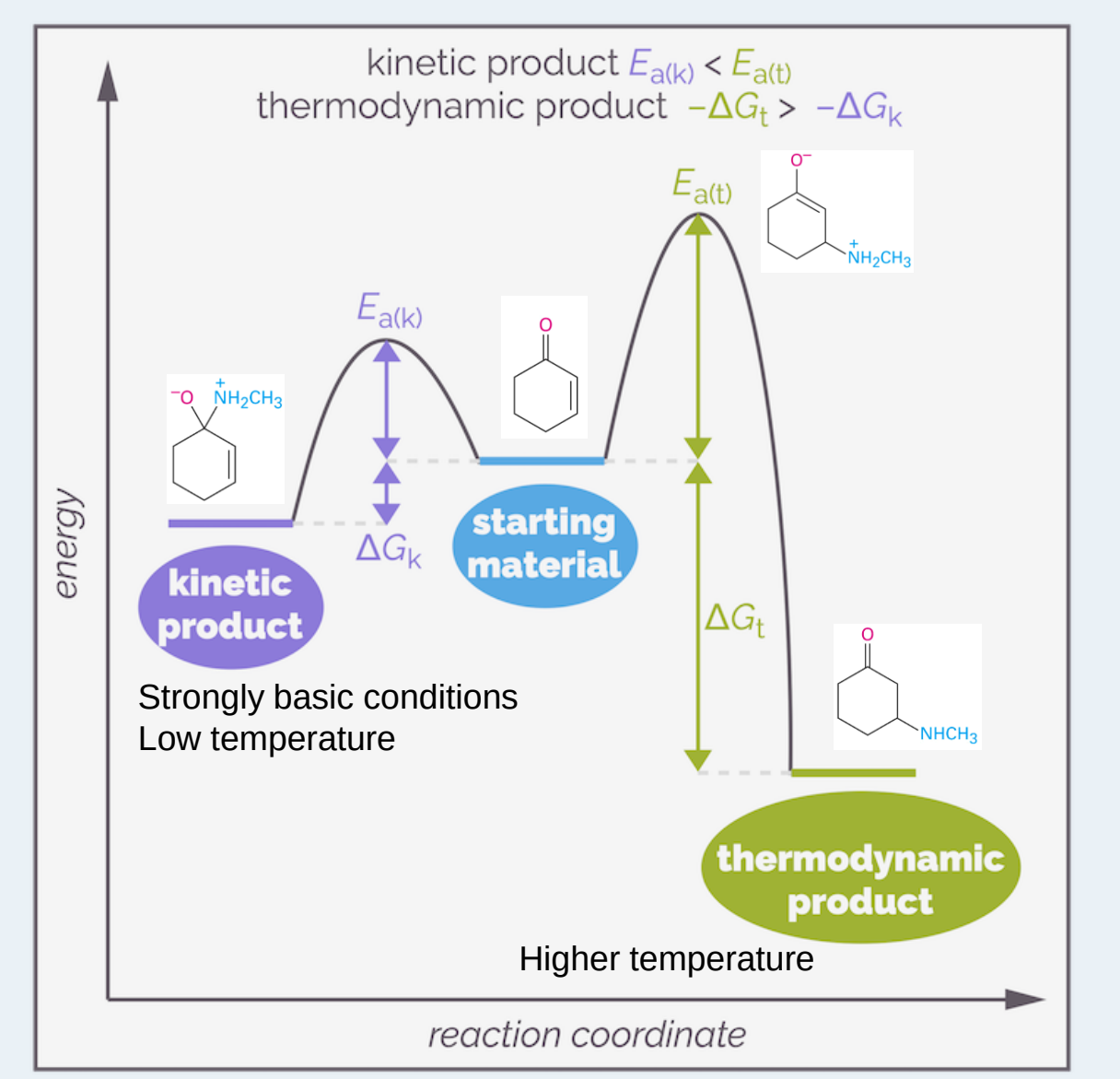
Chemoselectivity

1,2-addition vs Conjugate addition (1,4-addition)



https://www.makingmolecules.com/

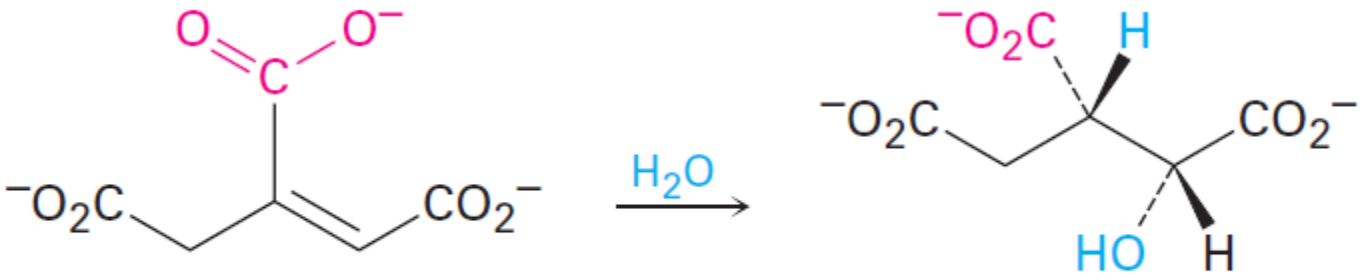
1,2-addition: lower activation barrier → kinetic control
1,4-addition: more stable product → thermodynamic control



https://www.makingmolecules.com/

Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

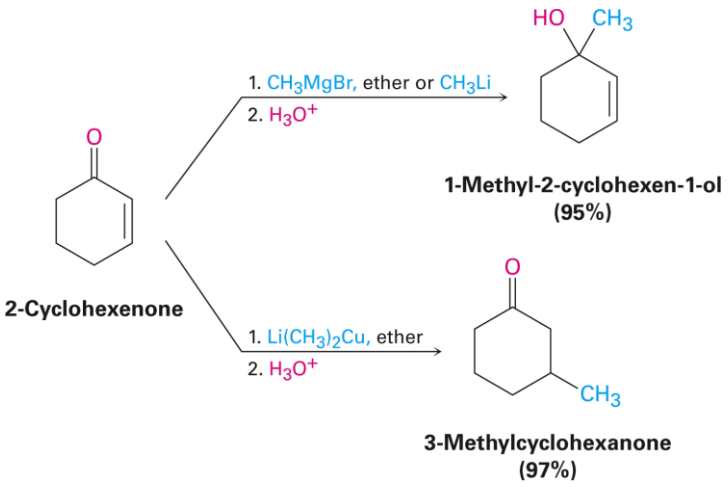
- Conjugate addition of water
  - Yields β-hydroxy aldehydes and ketones, by adding reversibly to α,β-unsaturated aldehydes and ketones
    - Position of the equilibrium generally favors unsaturated reactant



***cis*-Aconitate** **Isocitrate**

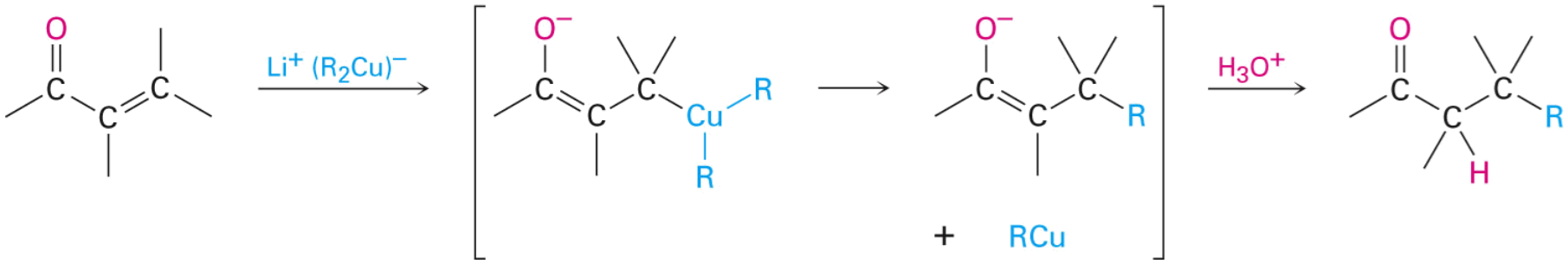
Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

- Organocopper Reactions
  - Reaction of an α,β-unsaturated ketone with a lithium diorganocopper reagent
    - Diorganocopper reagents form by reaction of 1 equivalent of cuprous iodide and 2 equivalents of organolithium
  - 1°, 2°, 3° alkyl, aryl, and alkenyl groups react
    - Alkynyl groups react poorly

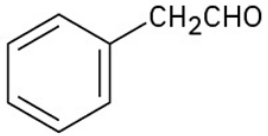


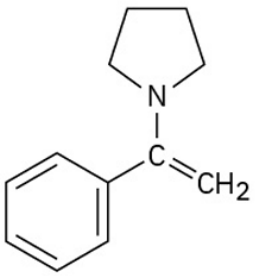
Conjugate Nucleophilic Addition to α,β-Unsaturated Aldehydes and Ketones

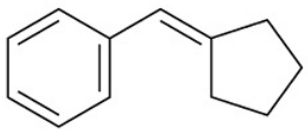
- Conjugate nucleophilic addition of a diorganocopper anion, R_2Cu^-, to a ketone
- Transfer of an R group and elimination of a neutral organocopper species, RCu, gives the final product



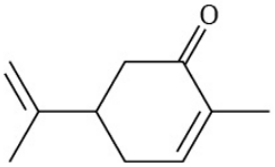
PROBLEM 19-62 How would you synthesize the following substances from benzaldehyde and any other reagents needed?

(a) 

(b) 

(c) 

PROBLEM 19-63 Carvone is the major constituent of spearmint oil. What products would you expect from reaction of carvone with the following reagents?

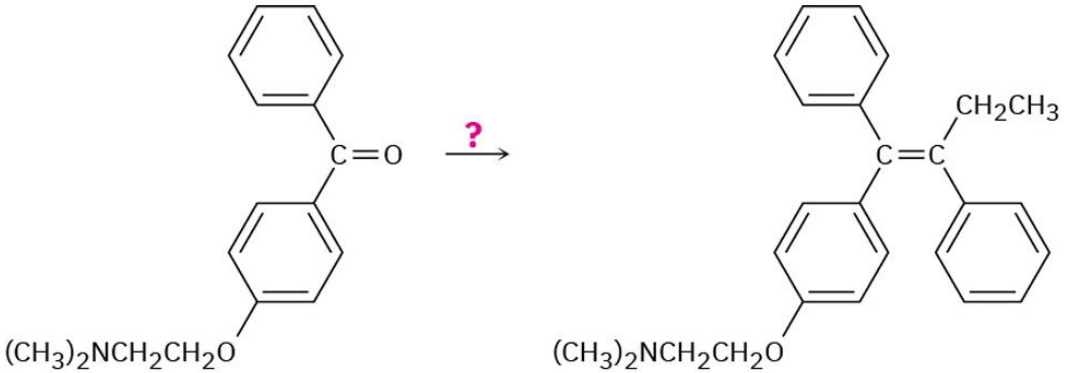


Carvone

(a) $(CH_3)_2Cu^-Li^+$, then H_3O^+ **(b)** $LiAlH_4$, then H_3O^+ **(c)** CH_3NH_2 **(d)** C_6H_5MgBr, then H_3O^+

(e) H_2/Pd **(f)** $HOCH_2CH_2OH$, HCl **(g)** $(C_6H_5)_3\overset{+}{P}\overset{-}{C}HCH_3$

PROBLEM 19-77 Tamoxifen is a drug used in the treatment of breast cancer. How would you prepare tamoxifen from benzene, the following ketone, and any other reagents needed?



Tamoxifen